

Unit 05: Utilities

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Objectives

After studying this unit, you will be able to:

- Understand the basic utilities
- Work with files
- Understand the Pipe
- Understand the compressing and archiving of files
- Understand the locating commands

Introduction

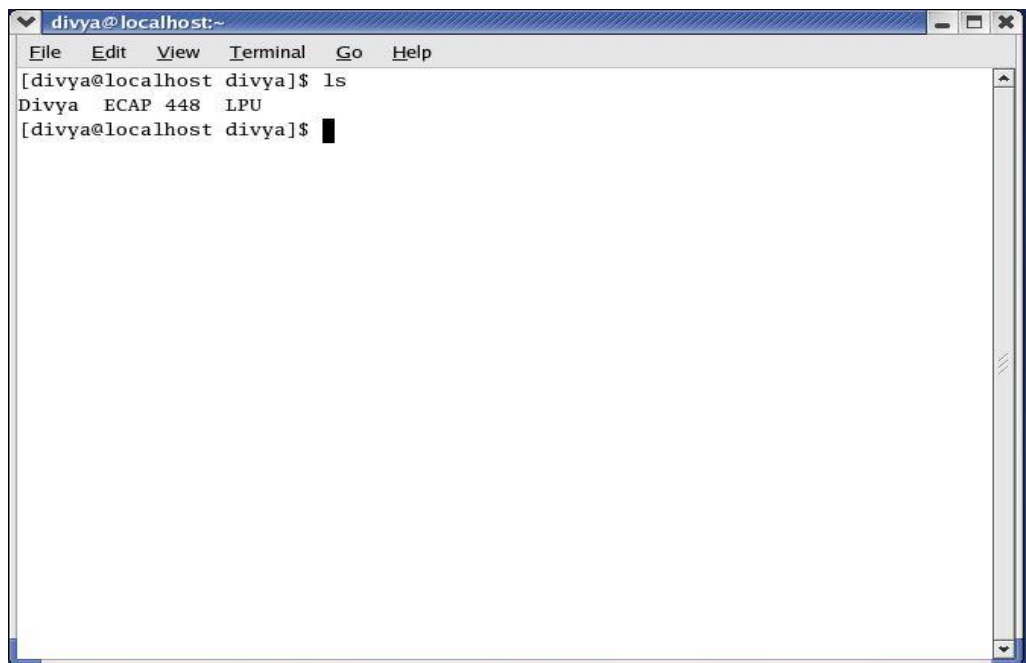
Command-line utilities are often faster, more powerful, or more complete than their GUI counterparts. When you work with a command-line interface, you are working with a shell. One of the important advantages of Linux is that it comes with thousands of utilities that perform innumerable functions.

- ls
- cat
- rm
- less
- more

5.1 Common Utilities

ls: Lists the Names of Files

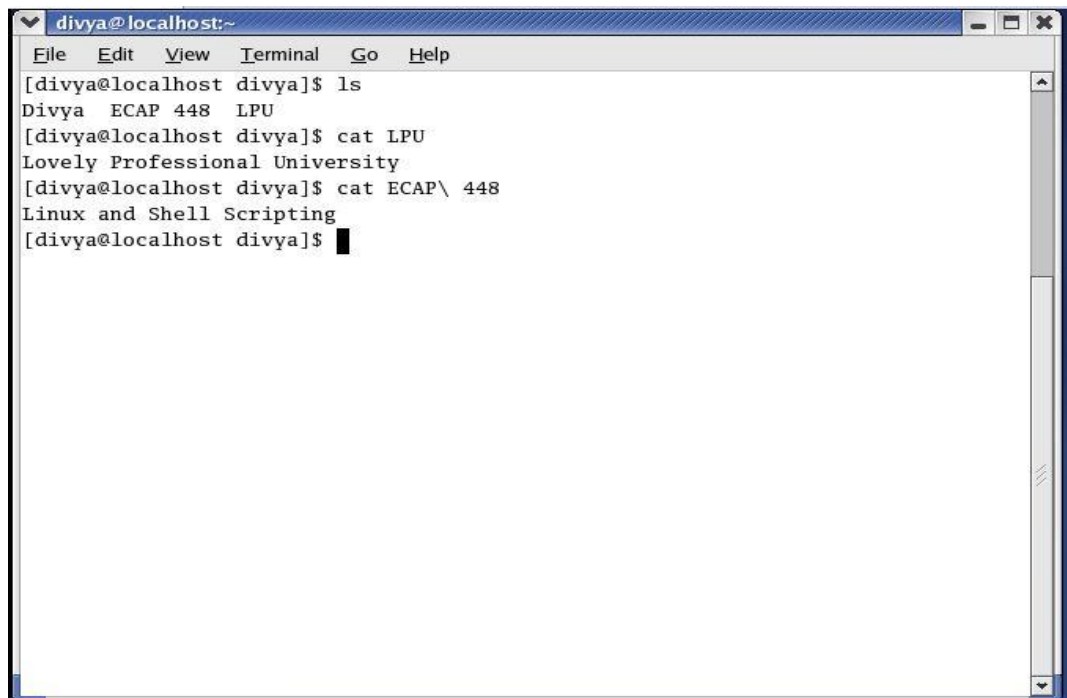
The ls utility lists the names of files which are available. ls is a Linux shell command that lists directory contents of files and directories.



```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
Divya ECAP 448 LPU  
[divya@localhost divya]$
```

cat: Displays the Text of File

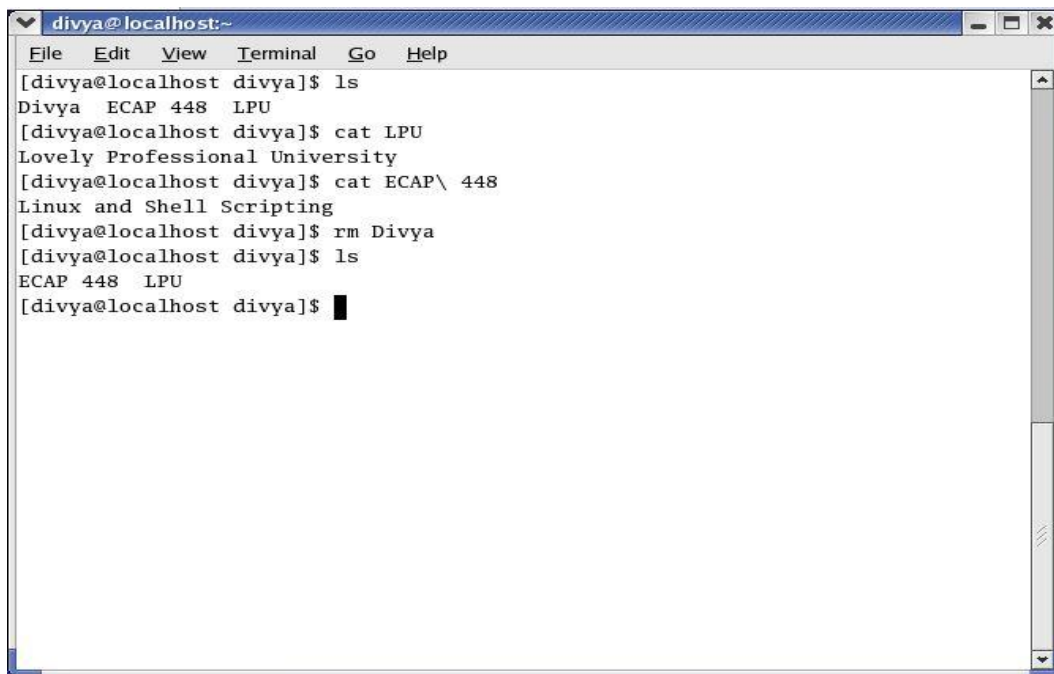
The cat utility displays the contents of a text file. The name of the command is derived from catenate, which means to join, one after the other.



```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
Divya ECAP 448 LPU  
[divya@localhost divya]$ cat LPU  
Lovely Professional University  
[divya@localhost divya]$ cat ECAP\ 448  
Linux and Shell Scripting  
[divya@localhost divya]$
```

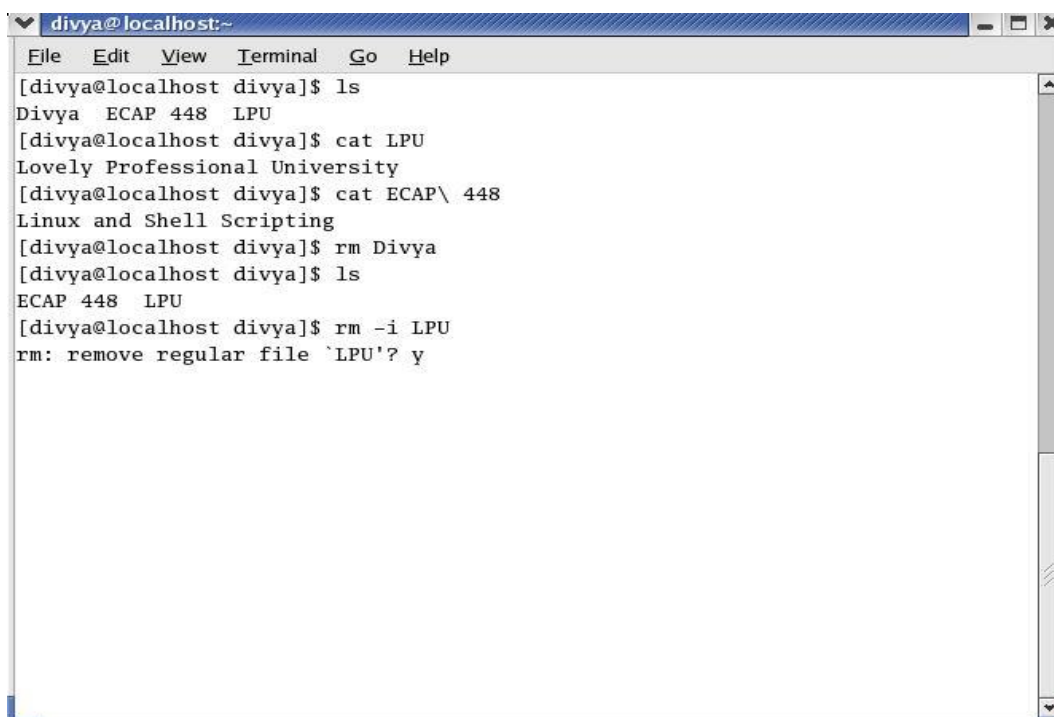
rm: Deletes a File

The rm (remove) utility deletes a file.

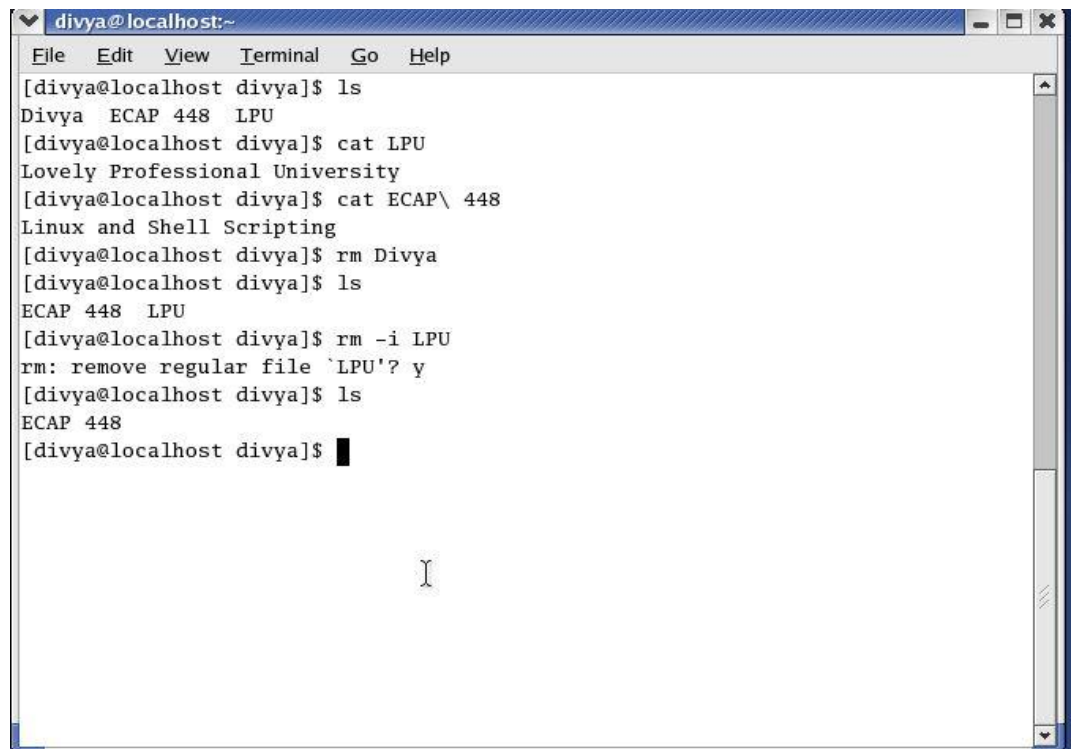


```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
Divya ECAP 448 LPU  
[divya@localhost divya]$ cat LPU  
Lovely Professional University  
[divya@localhost divya]$ cat ECAP\ 448  
Linux and Shell Scripting  
[divya@localhost divya]$ rm Divya  
[divya@localhost divya]$ ls  
ECAP 448 LPU  
[divya@localhost divya]$
```

When you follow `rm` with the `-i` option and the name of the file you want to delete, `rm` displays the name of the file and then waits for you to respond with `y` (yes) before it deletes the file. It does not delete the file if you respond with a string that begins with a character other than `y`.

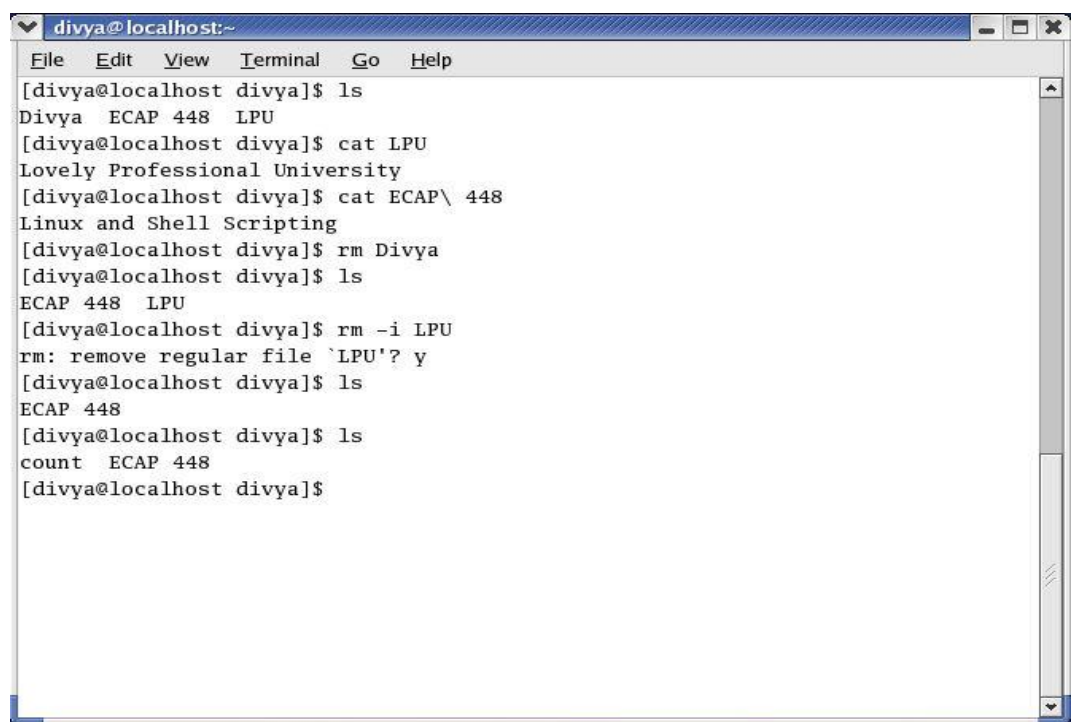


```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
Divya ECAP 448 LPU  
[divya@localhost divya]$ cat LPU  
Lovely Professional University  
[divya@localhost divya]$ cat ECAP\ 448  
Linux and Shell Scripting  
[divya@localhost divya]$ rm Divya  
[divya@localhost divya]$ ls  
ECAP 448 LPU  
[divya@localhost divya]$ rm -i LPU  
rm: remove regular file `LPU'? y
```

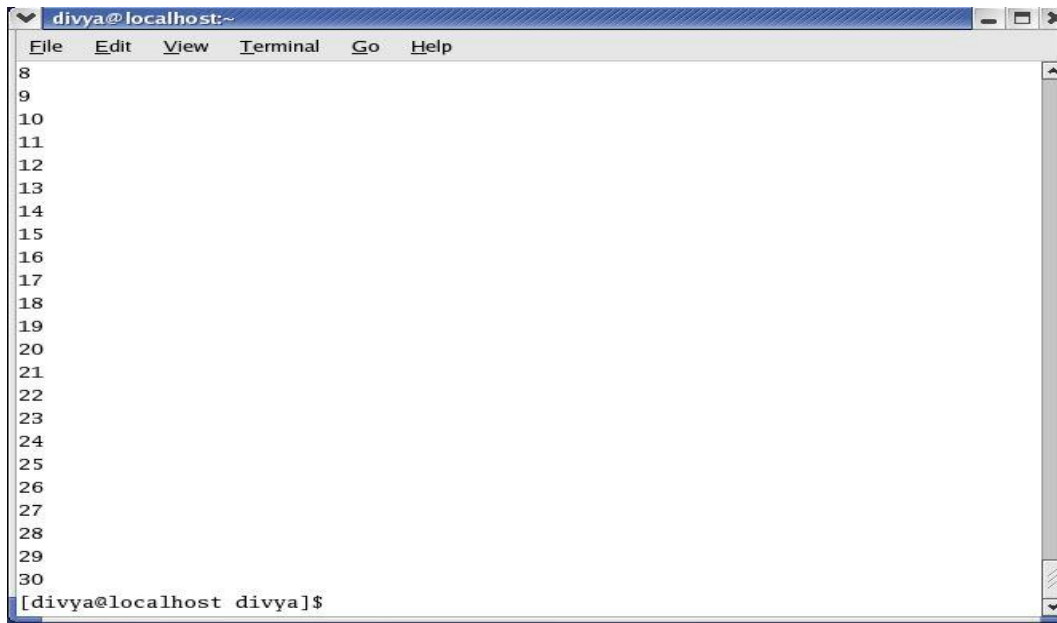
A terminal window titled 'divya@localhost:~' with a menu bar (File, Edit, View, Terminal, Go, Help). The terminal shows a series of commands and their outputs: 'ls' lists 'Divya', 'ECAP', '448', and 'LPU'; 'cat LPU' shows 'Lovely Professional University'; 'cat ECAP\ 448' shows 'Linux and Shell Scripting'; 'rm Divya' removes the file; 'ls' lists 'ECAP', '448', and 'LPU'; 'rm -i LPU' prompts 'rm: remove regular file `LPU'? y'; 'ls' lists 'ECAP' and '448'. The cursor is at the prompt '[divya@localhost divya]\$'.

less Is more: Display a Text File One Screen at a Time

It displays a text file one screen at a time. When you want to view a file that is longer than one screen, you can use either the less utility or the more utility. Each of these utilities pauses after displaying a screen of text; press the SPACE bar to display the next screen of text. Because these utilities show one page at a time, they are called pagers. Although less and more are very similar, they have subtle differences. At the end of the file, for example, less displays an END message and waits for you to press q before returning you to the shell. In contrast, more returns you directly to the shell. While using both utilities you can press h to display a Help screen that lists commands you can use while paging through a file.

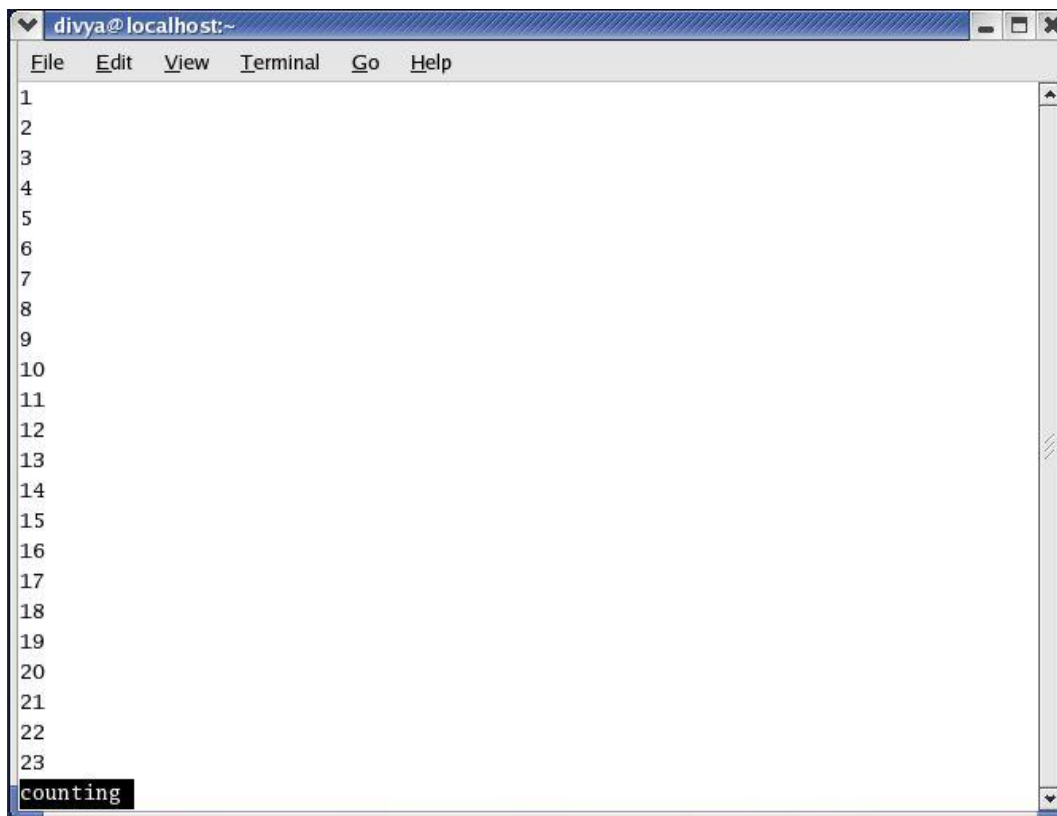
A terminal window titled 'divya@localhost:~' with a menu bar (File, Edit, View, Terminal, Go, Help). The terminal shows the same sequence of commands as the first image, but with an additional 'ls' command at the end, resulting in the output 'count ECAP 448'. The cursor is at the prompt '[divya@localhost divya]\$'.

Unit 05: Utilities



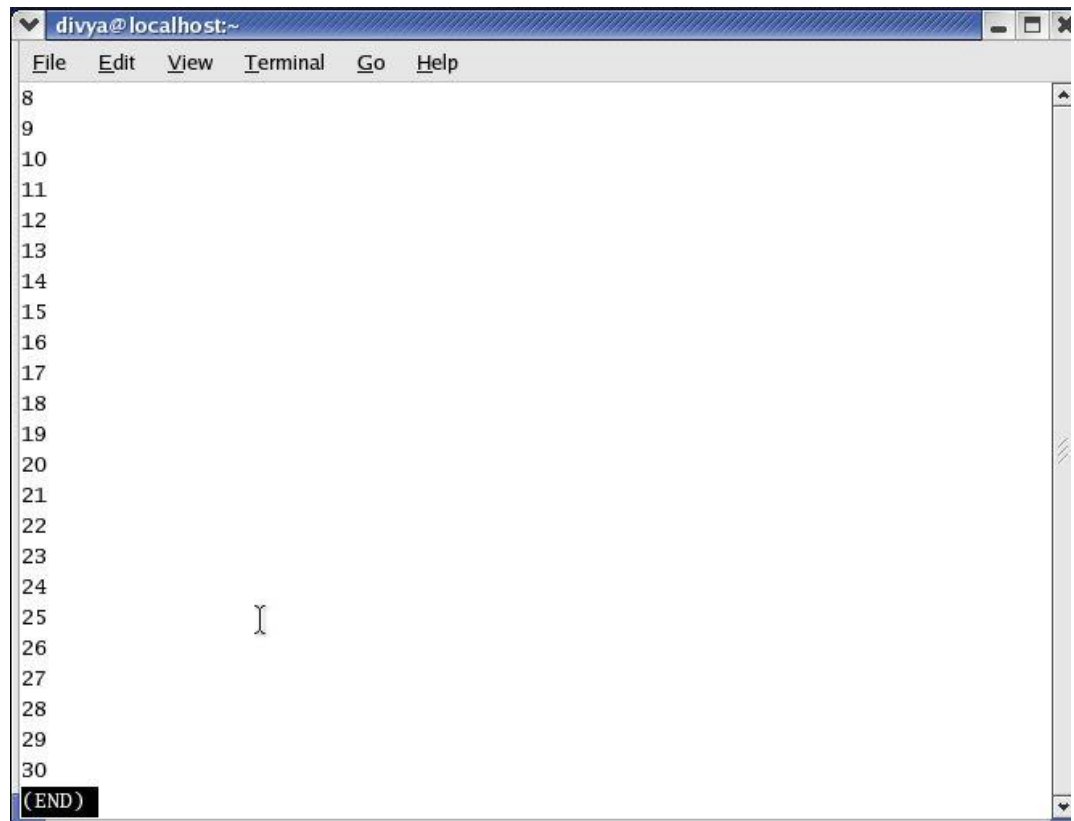
A terminal window titled "divya@localhost:~" with a menu bar containing "File", "Edit", "View", "Terminal", "Go", and "Help". The terminal displays a vertical list of numbers from 8 to 30. At the bottom, the prompt "[divya@localhost divya]" is followed by a dollar sign "\$".

```
divya@localhost:~  
File Edit View Terminal Go Help  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
24  
25  
26  
27  
28  
29  
30  
[divya@localhost divya]$
```

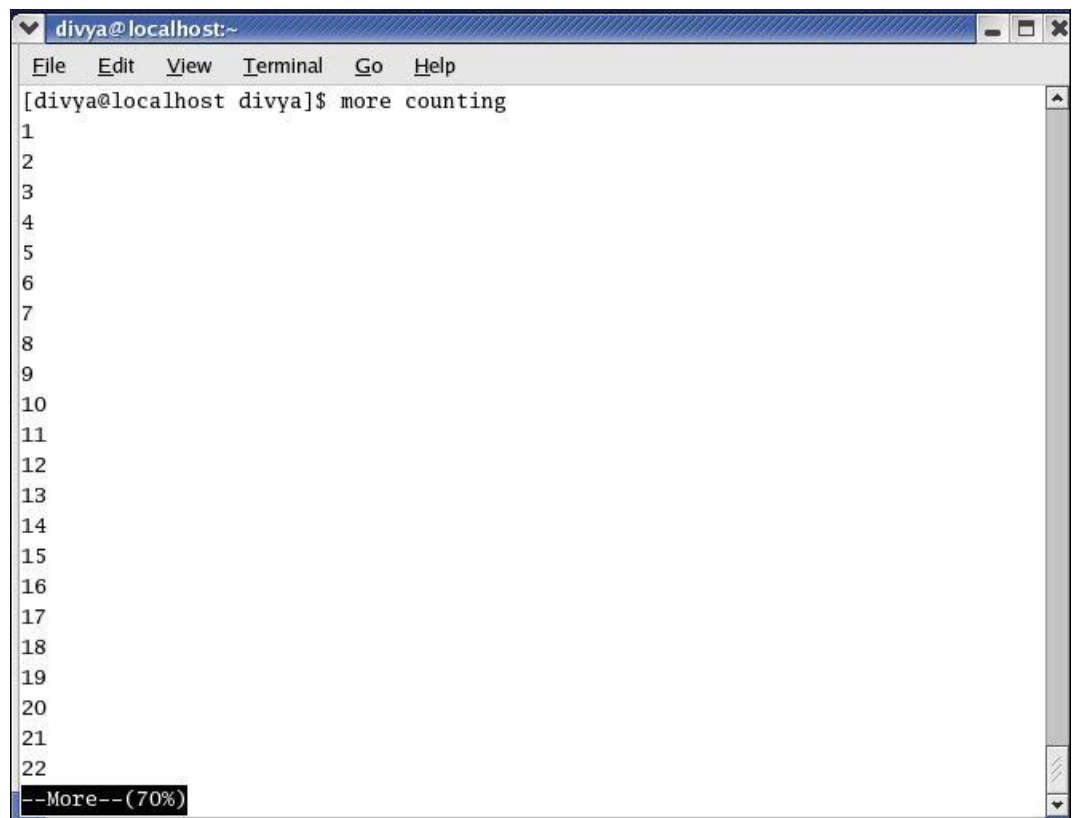


A terminal window titled "divya@localhost:~" with a menu bar containing "File", "Edit", "View", "Terminal", "Go", and "Help". The terminal displays a vertical list of numbers from 1 to 23. The word "counting" is highlighted in a black box at the bottom left of the terminal area. The prompt "[divya@localhost divya]" is followed by a dollar sign "\$".

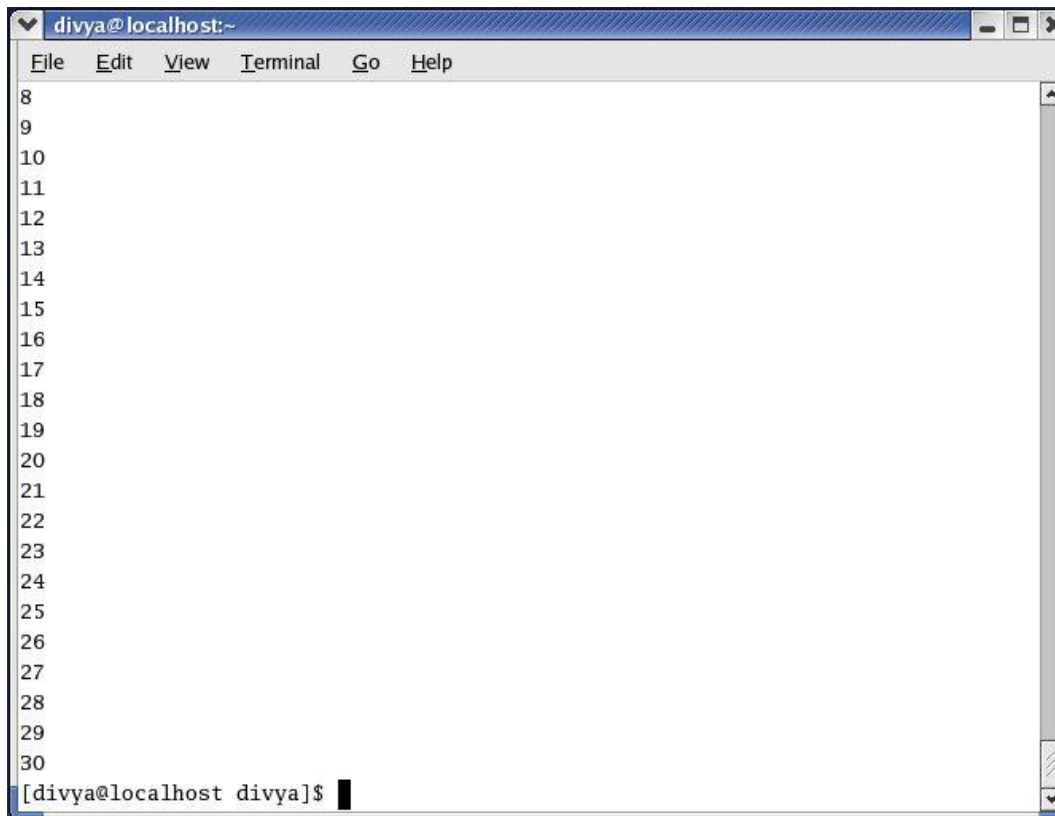
```
divya@localhost:~  
File Edit View Terminal Go Help  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
counting  
[divya@localhost divya]$
```



```
divya@localhost:~  
File Edit View Terminal Go Help  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
24  
25  
26  
27  
28  
29  
30  
(END)
```



```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ more counting  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
--More-- (70%)
```



```
divya@localhost:~
File Edit View Terminal Go Help
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
[divya@localhost divya]$
```

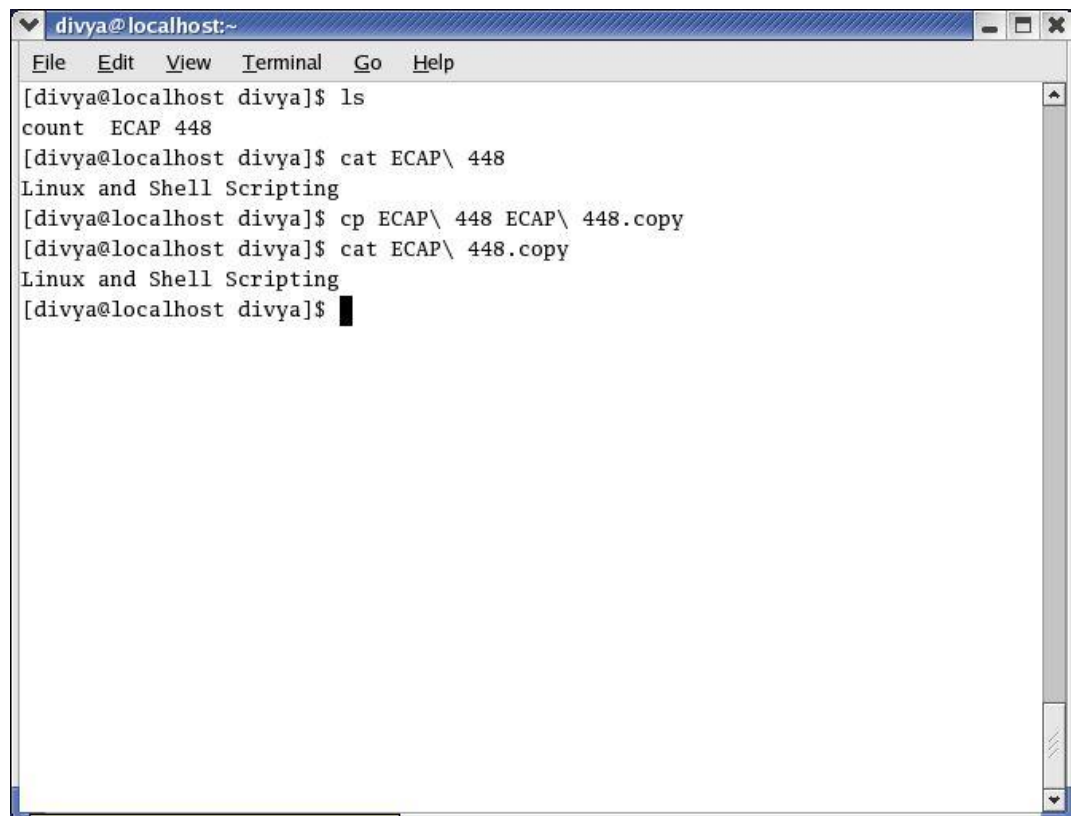
5.2 Working With Files

There are various utilities which we can use when we are working with files. These are:

- cp
- mv
- grep
- head
- tail
- sort
- uniq
- diff
- file

cp: Copies a File

The cp (copy) utility makes a copy of a file. This utility can copy any file, including text and executable program (binary) files. You can use cp to make a backup copy of a file or a copy to experiment with. The cp command line uses the following syntax to specify source and destination files: **cp source-file destination-file**

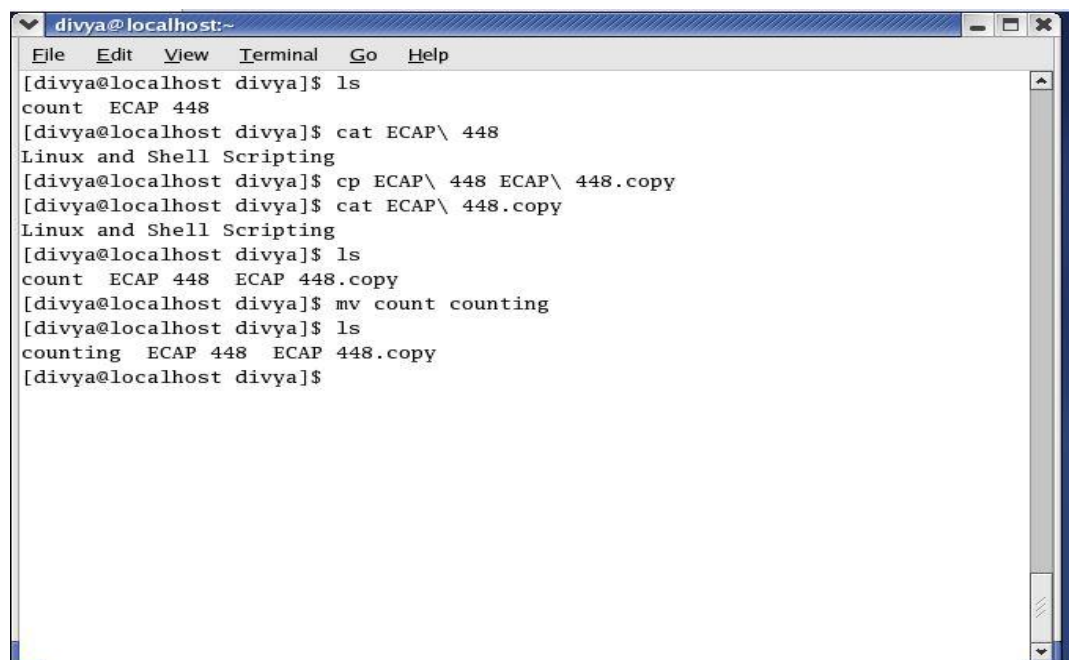


```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
count ECAP 448  
[divya@localhost divya]$ cat ECAP\ 448  
Linux and Shell Scripting  
[divya@localhost divya]$ cp ECAP\ 448 ECAP\ 448.copy  
[divya@localhost divya]$ cat ECAP\ 448.copy  
Linux and Shell Scripting  
[divya@localhost divya]$
```

cp can destroy a file: if the destination-file exists before you give a cp command, cp overwrites it. The cp -i (interactive) option prompts you before it overwrites a file.

mv: Changes the Name of a File

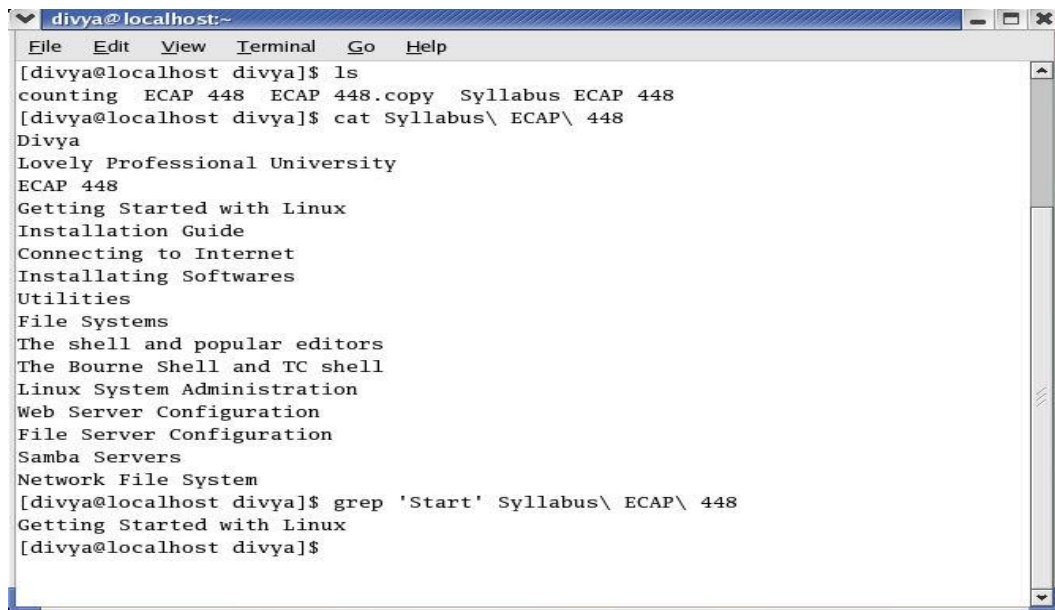
The mv (move) utility can rename a file without making a copy of it. The mv command line specifies an existing file and a new filename using the same syntax as cp: **mv existing-filename new-filename**



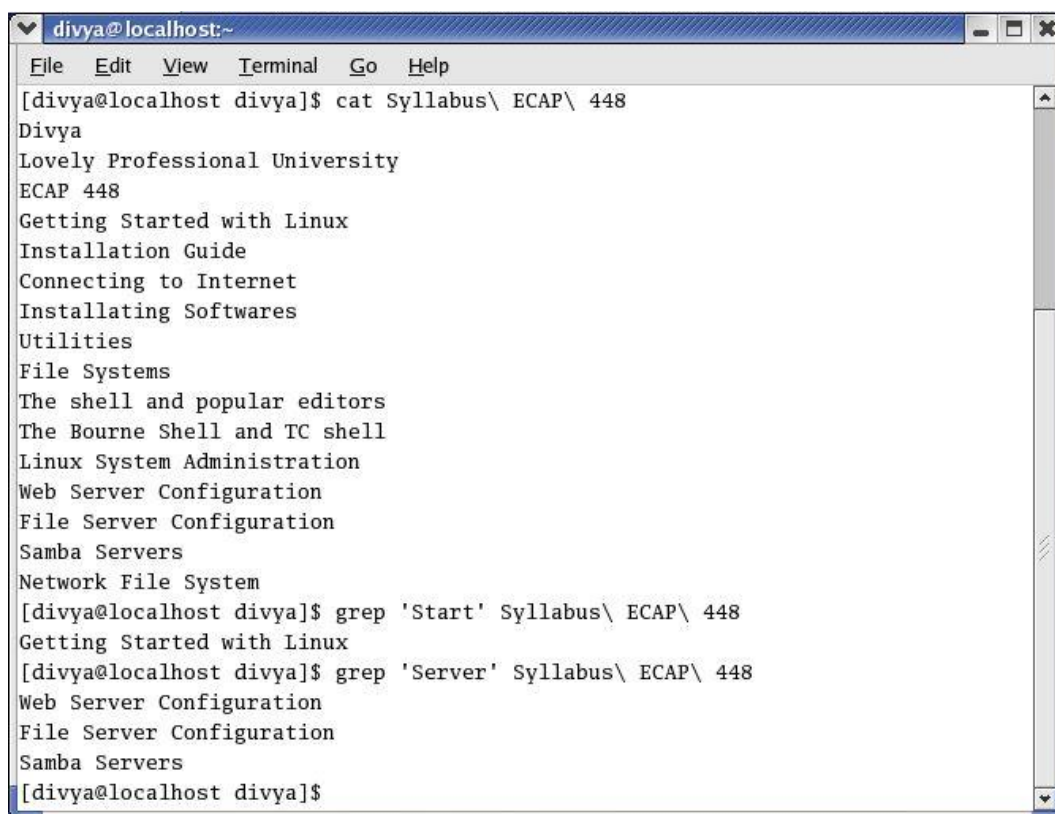
```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
count ECAP 448  
[divya@localhost divya]$ cat ECAP\ 448  
Linux and Shell Scripting  
[divya@localhost divya]$ cp ECAP\ 448 ECAP\ 448.copy  
[divya@localhost divya]$ cat ECAP\ 448.copy  
Linux and Shell Scripting  
[divya@localhost divya]$ ls  
count ECAP 448 ECAP 448.copy  
[divya@localhost divya]$ mv count counting  
[divya@localhost divya]$ ls  
counting ECAP 448 ECAP 448.copy  
[divya@localhost divya]$
```


grep: Searches for a String

The grep utility searches through one or more files to see whether any contain a specified string of characters. This utility does not change the file it searches but simply displays each line that contains the string.



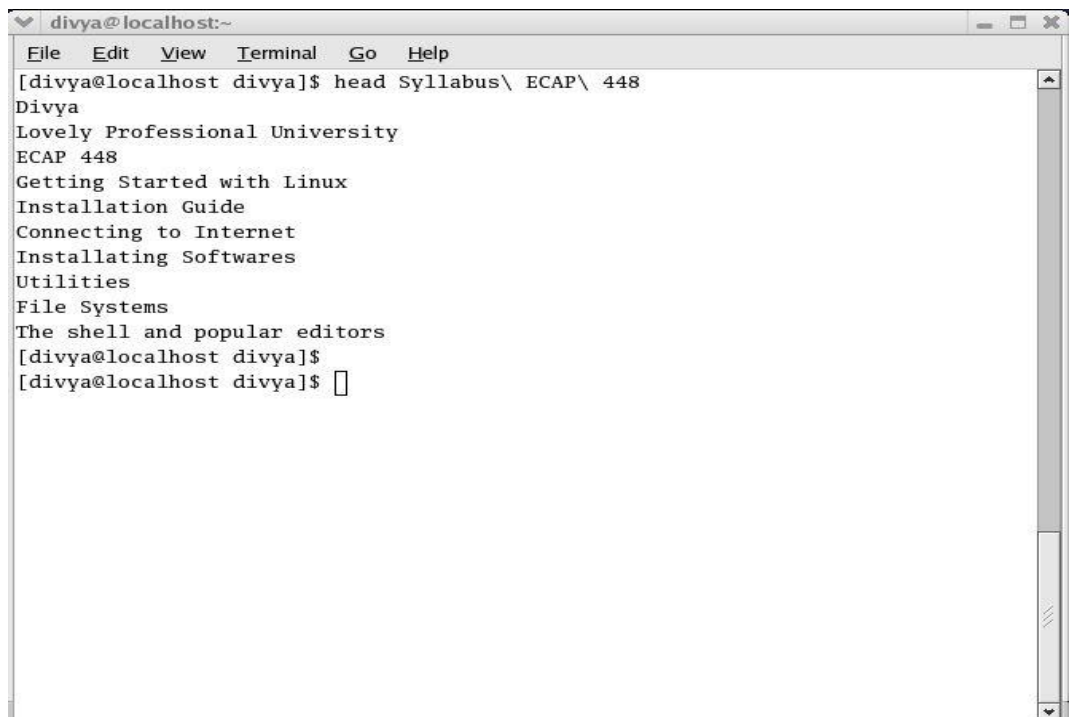
```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
counting ECAP 448 ECAP 448.copy Syllabus ECAP 448  
[divya@localhost divya]$ cat Syllabus\ ECAP\ 448  
Divya  
Lovely Professional University  
ECAP 448  
Getting Started with Linux  
Installation Guide  
Connecting to Internet  
Installing Softwares  
Utilities  
File Systems  
The shell and popular editors  
The Bourne Shell and TC shell  
Linux System Administration  
Web Server Configuration  
File Server Configuration  
Samba Servers  
Network File System  
[divya@localhost divya]$ grep 'Start' Syllabus\ ECAP\ 448  
Getting Started with Linux  
[divya@localhost divya]$
```



```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ cat Syllabus\ ECAP\ 448  
Divya  
Lovely Professional University  
ECAP 448  
Getting Started with Linux  
Installation Guide  
Connecting to Internet  
Installing Softwares  
Utilities  
File Systems  
The shell and popular editors  
The Bourne Shell and TC shell  
Linux System Administration  
Web Server Configuration  
File Server Configuration  
Samba Servers  
Network File System  
[divya@localhost divya]$ grep 'Start' Syllabus\ ECAP\ 448  
Getting Started with Linux  
[divya@localhost divya]$ grep 'Server' Syllabus\ ECAP\ 448  
Web Server Configuration  
File Server Configuration  
Samba Servers  
[divya@localhost divya]$
```

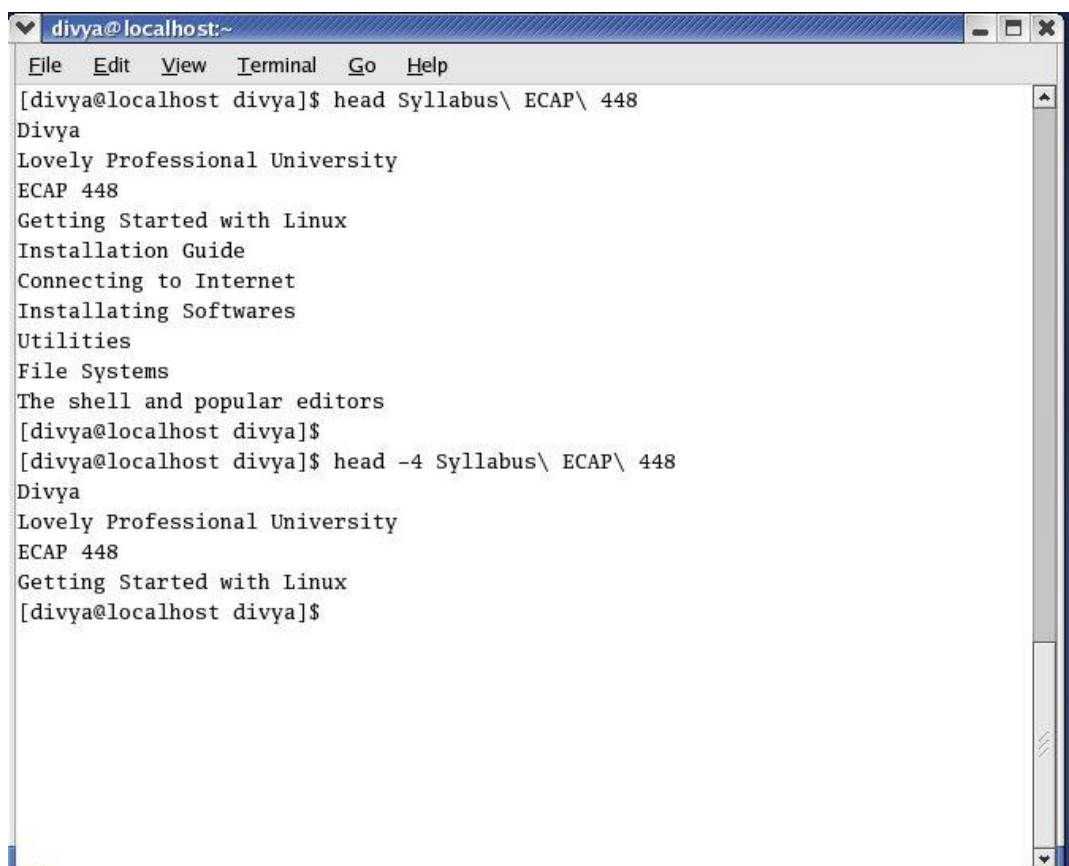
head: Displays the Beginning of a File

By default, the head utility displays the first ten lines of a file. For example, if you have a file named months that lists the 12 months of the year in calendar order, one to a line, then head displays Jan through Oct.



```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ head Syllabus\ ECAP\ 448  
Divya  
Lovely Professional University  
ECAP 448  
Getting Started with Linux  
Installation Guide  
Connecting to Internet  
Installing Softwares  
Utilities  
File Systems  
The shell and popular editors  
[divya@localhost divya]$  
[divya@localhost divya]$
```

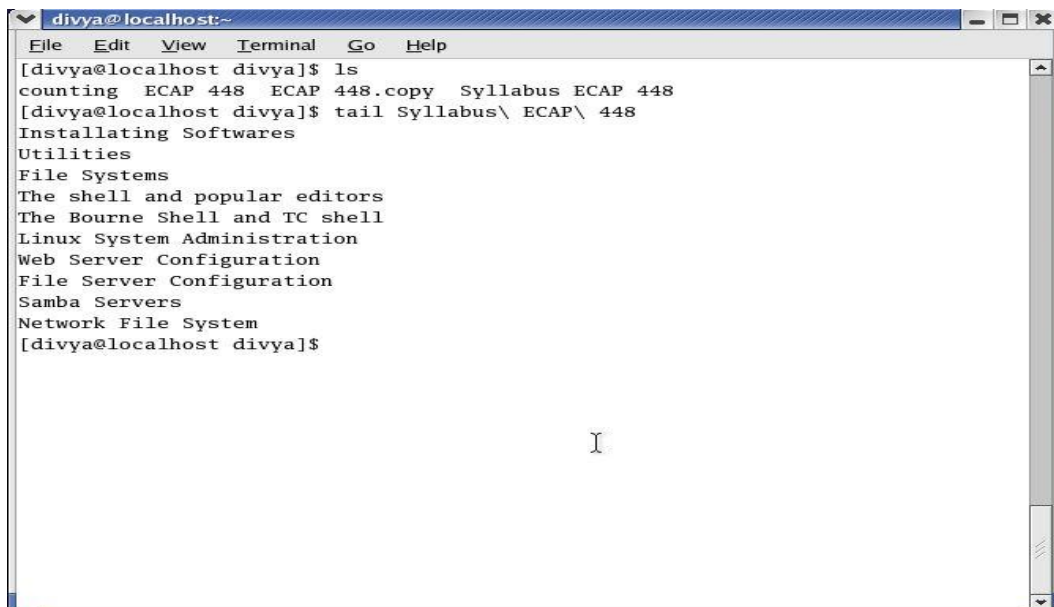
This utility can display any number of lines, so you can use it to look at only the first line of a file, at a full screen, or even more. To specify the number of lines displayed, include a hyphen followed by the number of lines you want head to display.



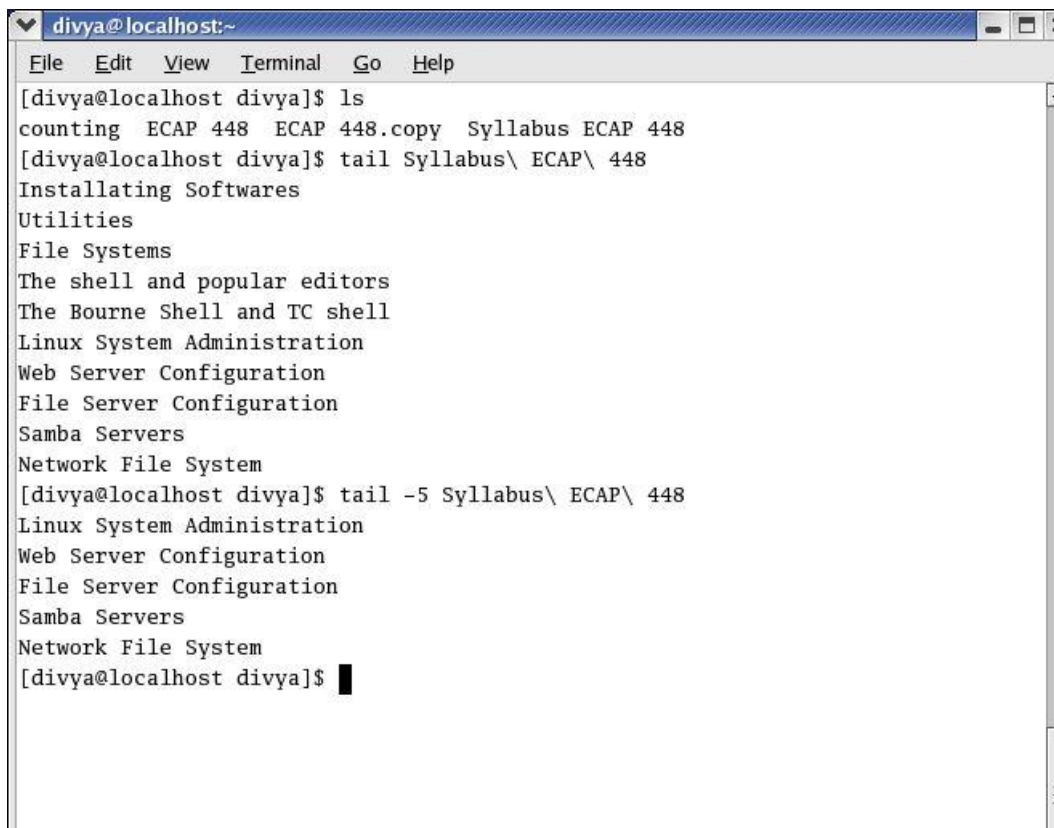
```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ head Syllabus\ ECAP\ 448  
Divya  
Lovely Professional University  
ECAP 448  
Getting Started with Linux  
Installation Guide  
Connecting to Internet  
Installing Softwares  
Utilities  
File Systems  
The shell and popular editors  
[divya@localhost divya]$  
[divya@localhost divya]$ head -4 Syllabus\ ECAP\ 448  
Divya  
Lovely Professional University  
ECAP 448  
Getting Started with Linux  
[divya@localhost divya]$
```

tail: Displays the End of a File

The tail utility is like head but by default displays the last ten lines of a file. Depending on how you invoke it, this utility can display fewer or more than ten lines.



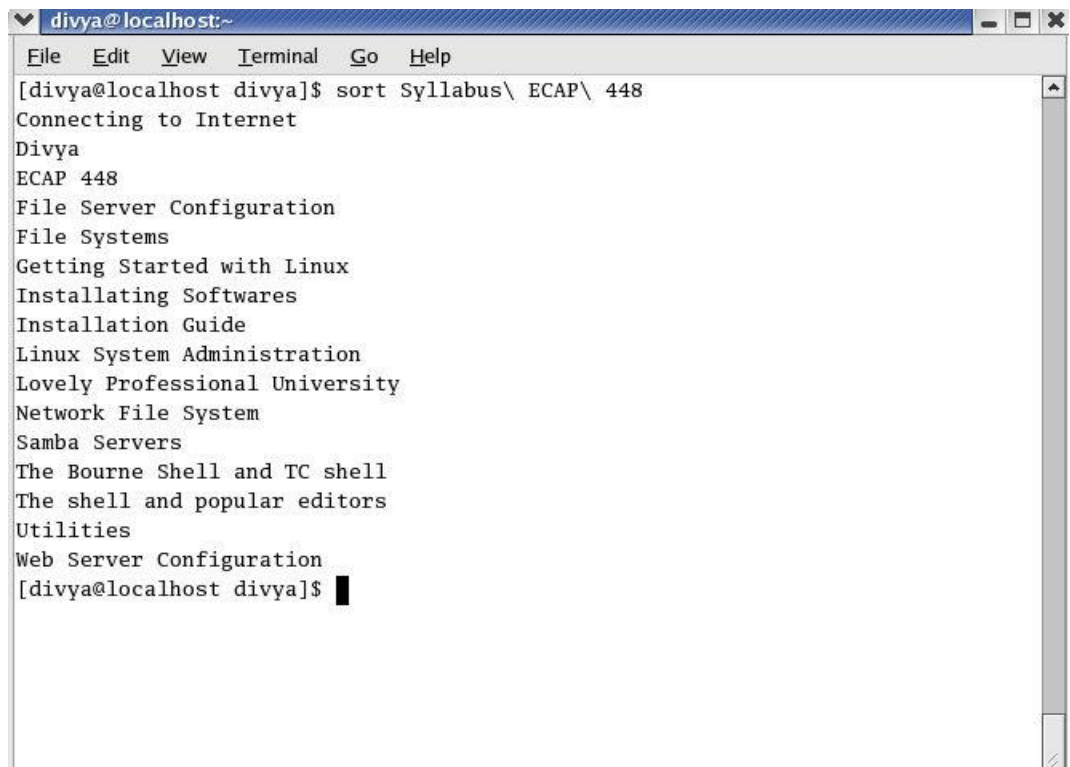
```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
counting ECAP 448 ECAP 448.copy Syllabus ECAP 448  
[divya@localhost divya]$ tail Syllabus\ ECAP\ 448  
Installing Softwares  
Utilities  
File Systems  
The shell and popular editors  
The Bourne Shell and TC shell  
Linux System Administration  
Web Server Configuration  
File Server Configuration  
Samba Servers  
Network File System  
[divya@localhost divya]$
```



```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
counting ECAP 448 ECAP 448.copy Syllabus ECAP 448  
[divya@localhost divya]$ tail Syllabus\ ECAP\ 448  
Installing Softwares  
Utilities  
File Systems  
The shell and popular editors  
The Bourne Shell and TC shell  
Linux System Administration  
Web Server Configuration  
File Server Configuration  
Samba Servers  
Network File System  
[divya@localhost divya]$ tail -5 Syllabus\ ECAP\ 448  
Linux System Administration  
Web Server Configuration  
File Server Configuration  
Samba Servers  
Network File System  
[divya@localhost divya]$
```

sort: Displays a File in Order

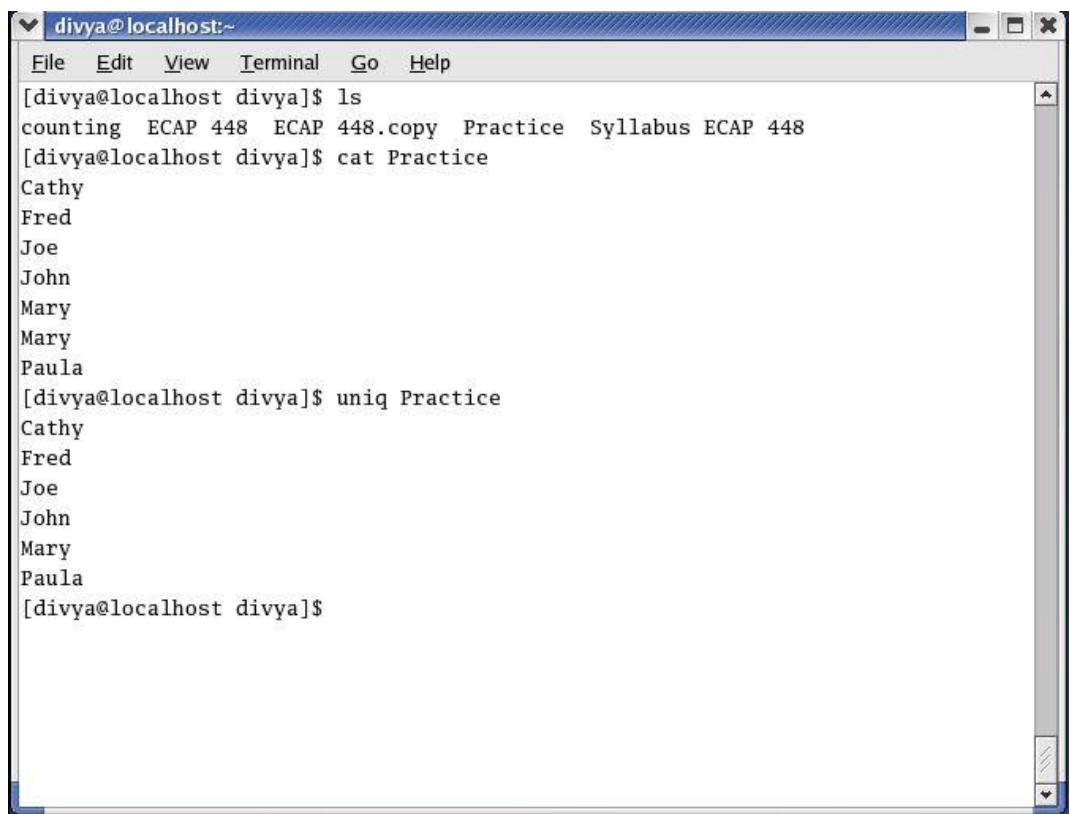
The sort utility displays the contents of a file in order by lines; it does not change the original file.



```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ sort Syllabus\ ECAP\ 448  
Connecting to Internet  
Divya  
ECAP 448  
File Server Configuration  
File Systems  
Getting Started with Linux  
Installing Softwares  
Installation Guide  
Linux System Administration  
Lovely Professional University  
Network File System  
Samba Servers  
The Bourne Shell and TC shell  
The shell and popular editors  
Utilities  
Web Server Configuration  
[divya@localhost divya]$
```

uniq: Removes Duplicate Lines from a File

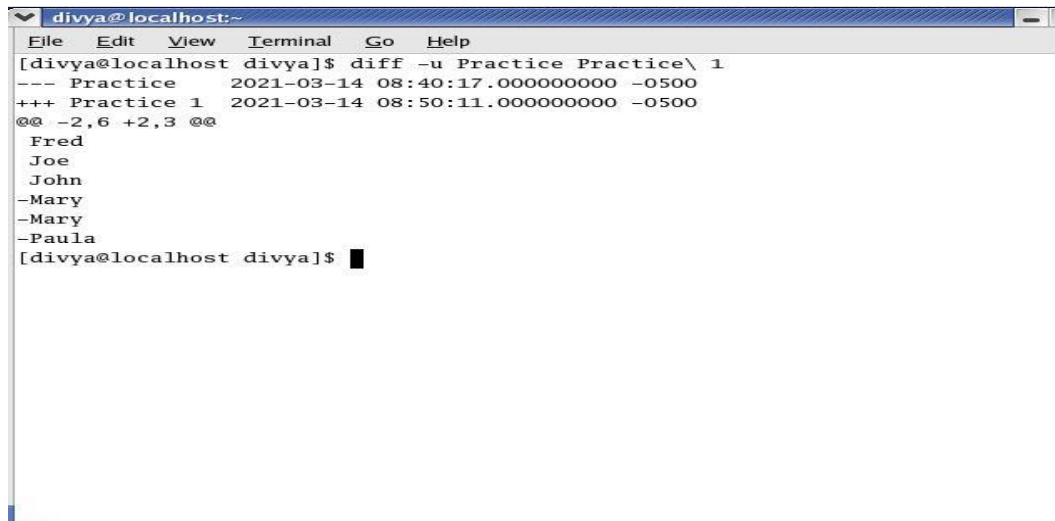
The `uniq` (unique) utility displays a file, skipping adjacent duplicate lines, but does not change the original file. If a file contains a list of names and has two successive entries for the same person, `uniq` skips the duplicate line. If a file is sorted before it is processed by `uniq`, this utility ensures that no two lines in the file are the same.



```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
counting ECAP 448 ECAP 448.copy Practice Syllabus ECAP 448  
[divya@localhost divya]$ cat Practice  
Cathy  
Fred  
Joe  
John  
Mary  
Mary  
Paula  
[divya@localhost divya]$ uniq Practice  
Cathy  
Fred  
Joe  
John  
Mary  
Paula  
[divya@localhost divya]$
```

diff: Compares Two Files

The diff (difference) utility compares two files and displays a list of the differences between them. This utility does not change either file; it is useful when you want to compare two versions of a letter or a report or two versions of the source code for a program. The diff utility with the -u (unified output format) option first displays two lines indicating which of the files you are comparing will be denoted by a plus sign (+) and which by a minus sign (-).



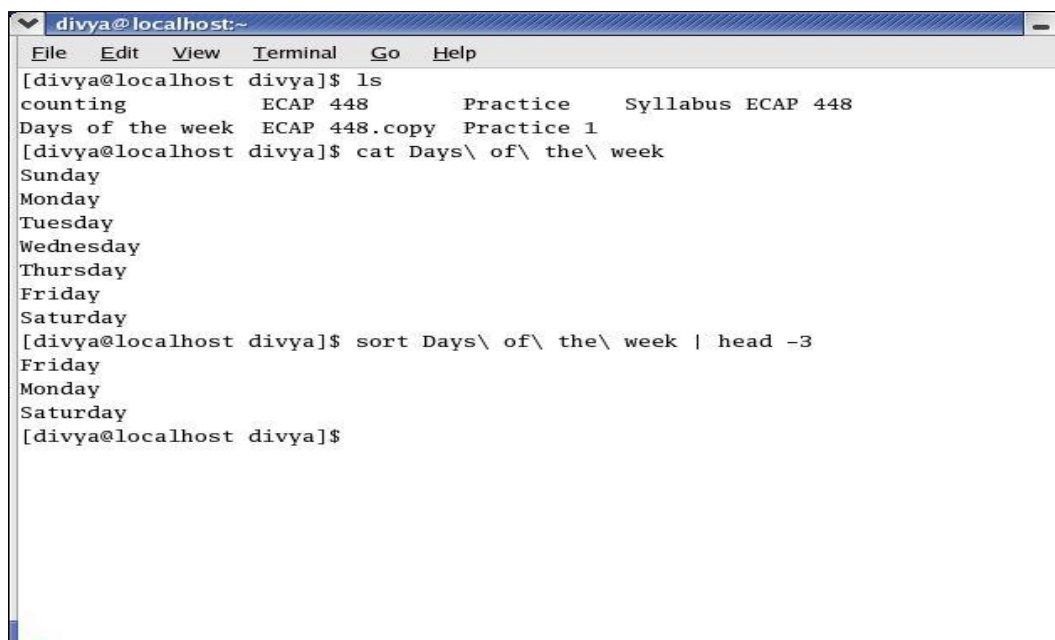
```
divya@localhost:~
File Edit View Terminal Go Help
[divya@localhost divya]$ diff -u Practice Practice\ 1
--- Practice      2021-03-14 08:40:17.000000000 -0500
+++ Practice 1    2021-03-14 08:50:11.000000000 -0500
@@ -2,6 +2,3 @@
 Fred
 Joe
 John
-Mary
-Mary
-Paula
[divya@localhost divya]$
```

file: Identifies the Contents of a File

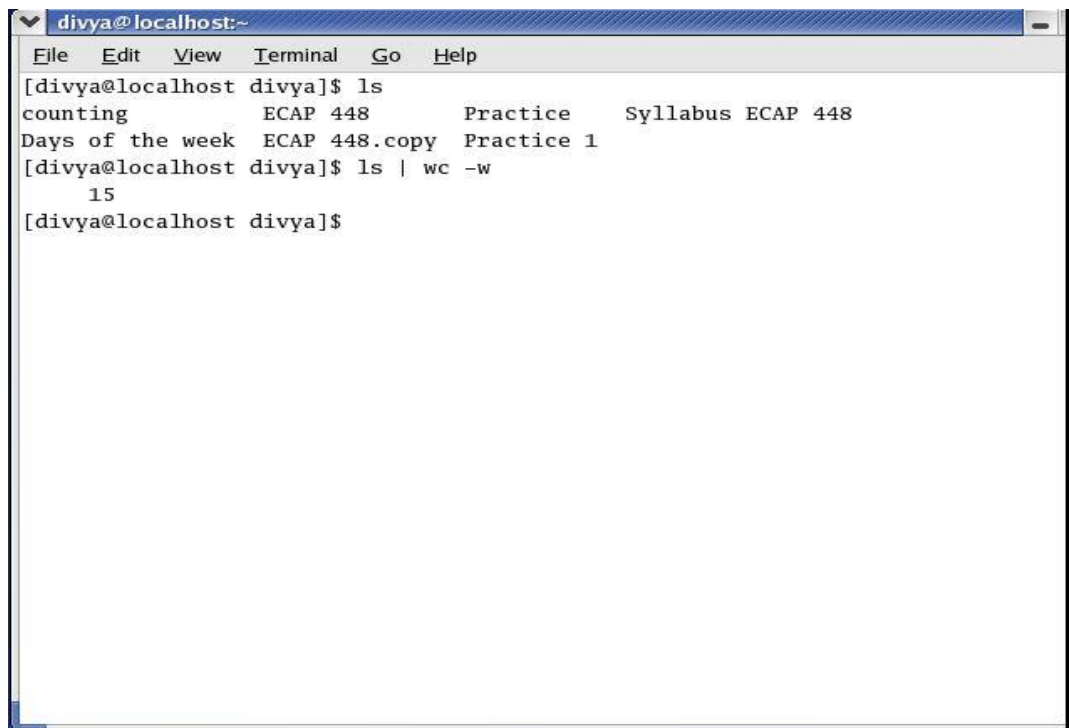
You can use the file utility to learn about the contents of a file without having to open and examine the file yourself.

| (Pipe): Communicates Between Processes

A process is the execution of a command by Linux. Communication between processes is one of the hallmarks of both UNIX and Linux. A pipe (written as a vertical bar [|] on the command line and appearing as a solid or broken vertical line on a keyboard) provides the simplest form of this kind of communication. Simply put, a pipe takes the output of one utility and sends that output as input to another utility.



```
divya@localhost:~
File Edit View Terminal Go Help
[divya@localhost divya]$ ls
counting          ECAP 448          Practice          Syllabus ECAP 448
Days of the week  ECAP 448.copy    Practice 1
[divya@localhost divya]$ cat Days\ of\ the\ week
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
[divya@localhost divya]$ sort Days\ of\ the\ week | head -3
Friday
Monday
Saturday
[divya@localhost divya]$
```



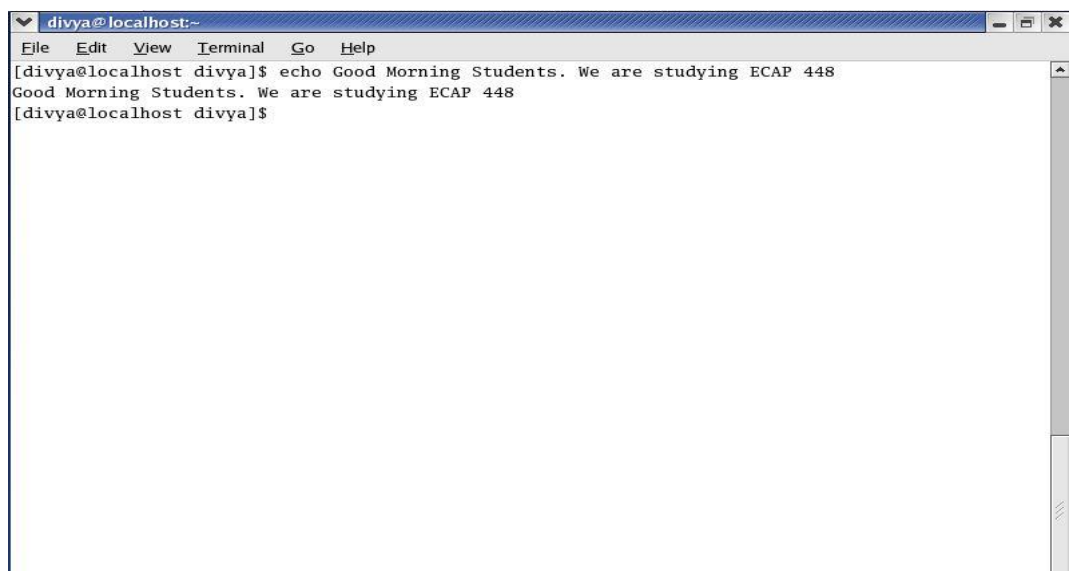
```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ ls  
counting          ECAP 448          Practice          Syllabus ECAP 448  
Days of the week  ECAP 448.copy    Practice 1  
[divya@localhost divya]$ ls | wc -w  
15  
[divya@localhost divya]$
```

5.3 Four More Utilities

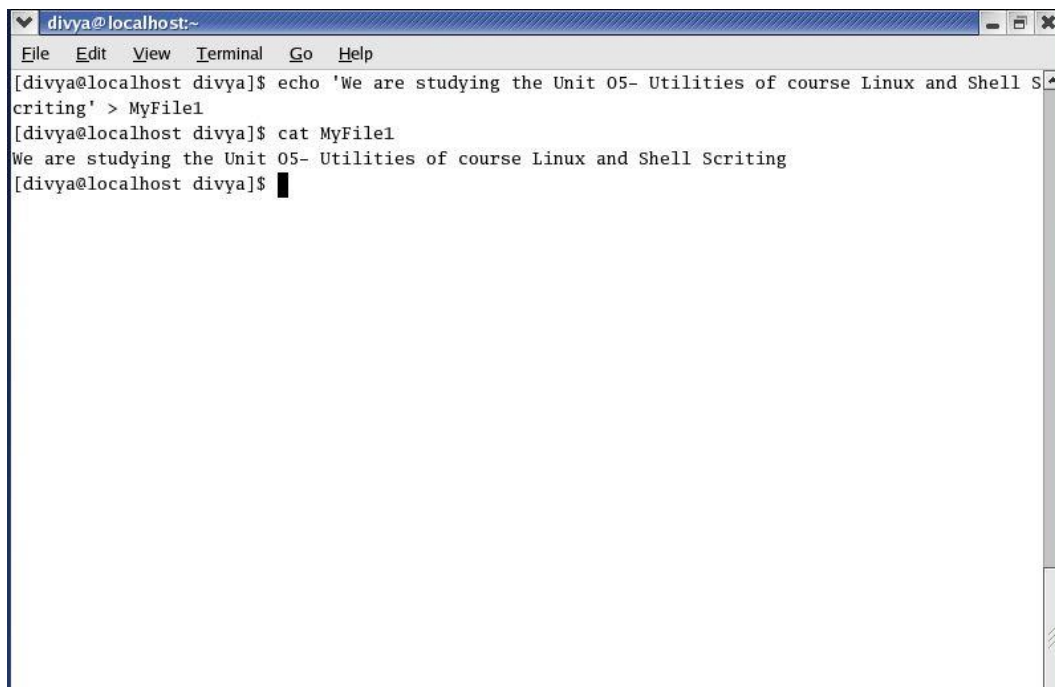
- echo
- date
- script
- unix2dos

echo: Displays Text

The echo utility copies the characters you type on the command line after echo to the screen. You can also send messages from shell scripts to the screen.



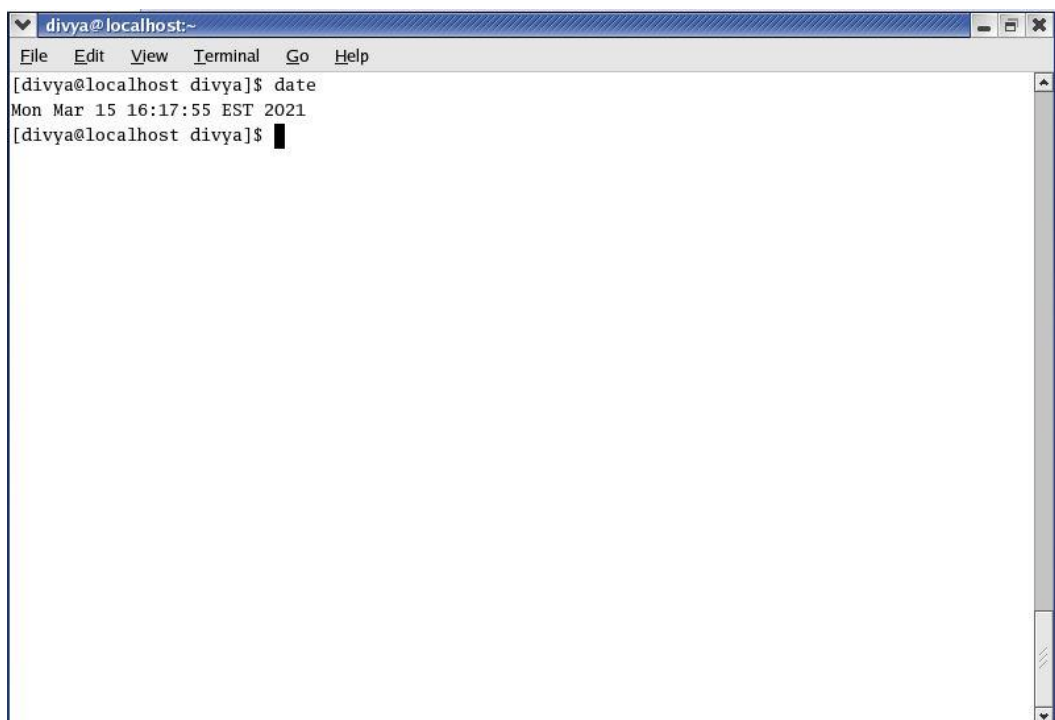
```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ echo Good Morning Students. We are studying ECAP 448  
Good Morning Students. We are studying ECAP 448  
[divya@localhost divya]$
```


A terminal window titled 'divya@localhost:~' with a menu bar (File, Edit, View, Terminal, Go, Help). The terminal shows the following commands and output:

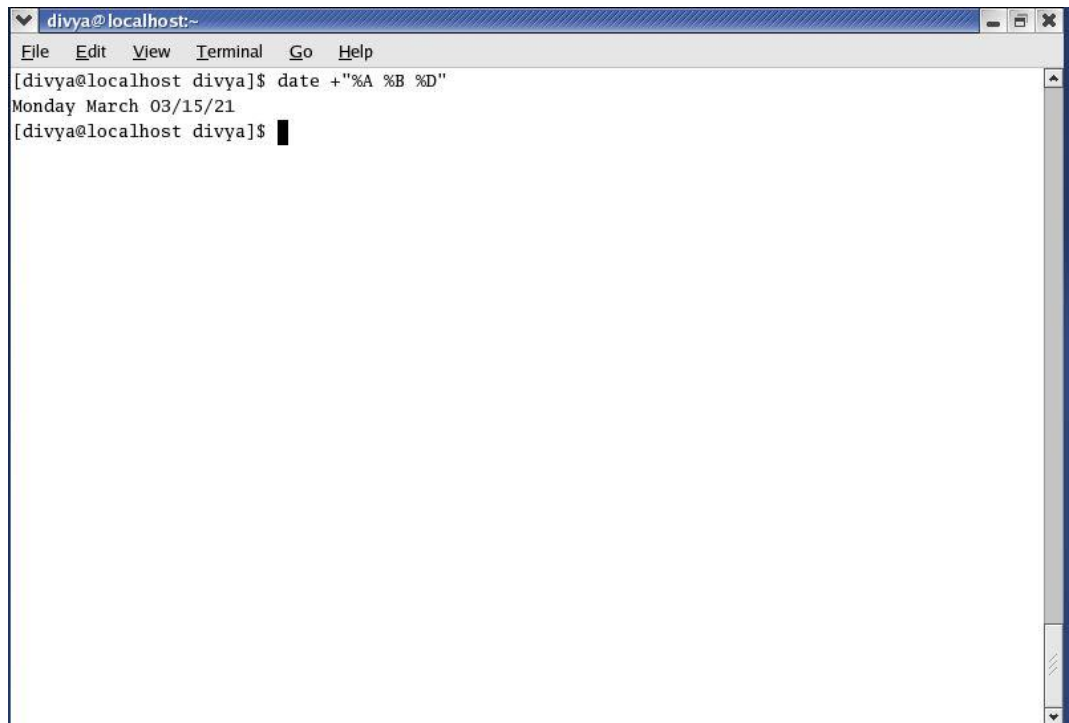
```
[divya@localhost divya]$ echo 'We are studying the Unit 05- Utilities of course Linux and Shell Scriting' > MyFile1
[divya@localhost divya]$ cat MyFile1
We are studying the Unit 05- Utilities of course Linux and Shell Scriting
[divya@localhost divya]$
```

date: Displays the Time and Date

The date utility displays the current date and time. You can choose the format and select the contents of the output of date.

A terminal window titled 'divya@localhost:~' with a menu bar (File, Edit, View, Terminal, Go, Help). The terminal shows the following command and output:

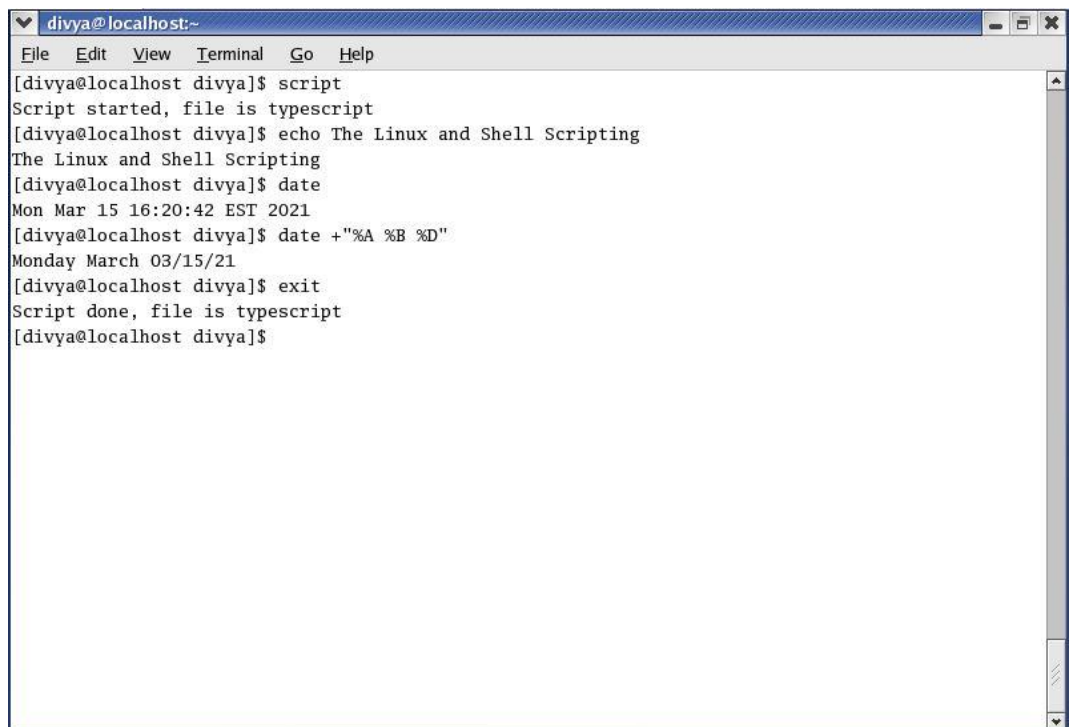
```
[divya@localhost divya]$ date
Mon Mar 15 16:17:55 EST 2021
[divya@localhost divya]$
```

A terminal window titled 'divya@localhost:~' with a menu bar (File, Edit, View, Terminal, Go, Help). The prompt is '[divya@localhost divya]\$'. The user enters 'date +%A %B %D' and the output is 'Monday March 03/15/21'. The prompt returns to '[divya@localhost divya]\$' with a cursor.

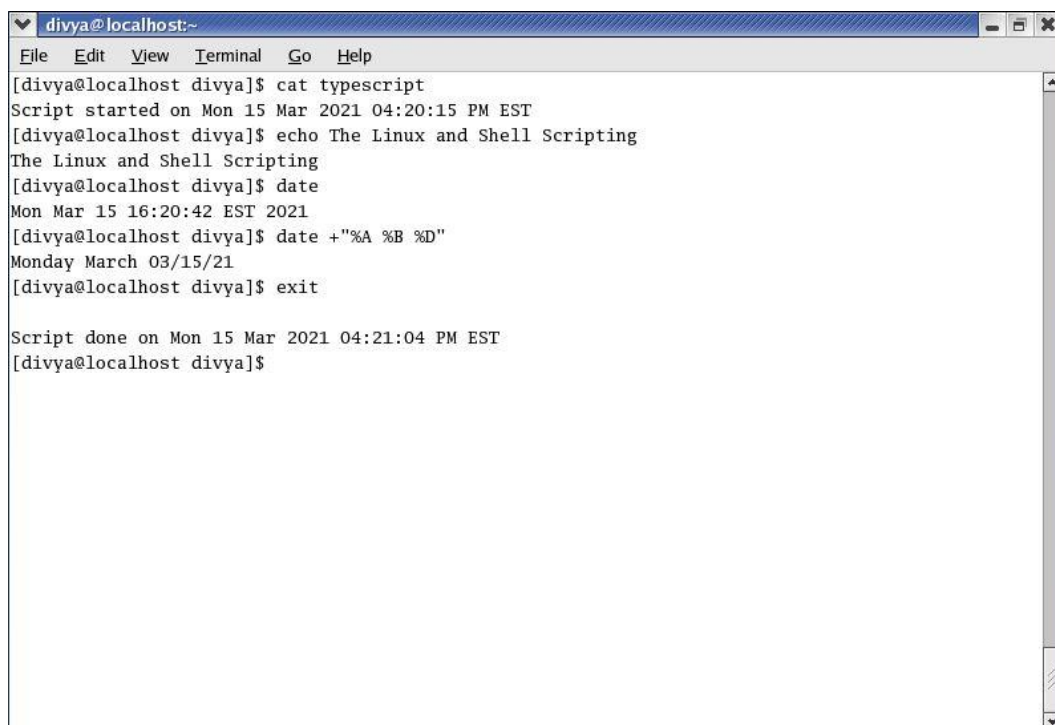
```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ date +%A %B %D  
Monday March 03/15/21  
[divya@localhost divya]$
```

script: Records a Shell Session

The script utility records all or part of a login session, including your input and the system's responses. This utility is useful only from character-based devices. By default, script captures the session in a file named typescript. To specify a different filename, follow the script command with a SPACE and the filename

A terminal window titled 'divya@localhost:~' with a menu bar (File, Edit, View, Terminal, Go, Help). The prompt is '[divya@localhost divya]\$'. The user enters 'script', and the output is 'Script started, file is typescript'. The user enters 'echo The Linux and Shell Scripting', and the output is 'The Linux and Shell Scripting'. The user enters 'date', and the output is 'Mon Mar 15 16:20:42 EST 2021'. The user enters 'date +%A %B %D', and the output is 'Monday March 03/15/21'. The user enters 'exit', and the output is 'Script done, file is typescript'. The prompt returns to '[divya@localhost divya]\$' with a cursor.

```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ script  
Script started, file is typescript  
[divya@localhost divya]$ echo The Linux and Shell Scripting  
The Linux and Shell Scripting  
[divya@localhost divya]$ date  
Mon Mar 15 16:20:42 EST 2021  
[divya@localhost divya]$ date +%A %B %D  
Monday March 03/15/21  
[divya@localhost divya]$ exit  
Script done, file is typescript  
[divya@localhost divya]$
```


A screenshot of a terminal window titled 'divya@localhost:~'. The window has a menu bar with 'File', 'Edit', 'View', 'Terminal', 'Go', and 'Help'. The terminal shows the execution of a shell script named 'typescript'. The script starts with a timestamp 'Mon 15 Mar 2021 04:20:15 PM EST'. It then echoes 'The Linux and Shell Scripting'. The user runs 'date', which outputs 'Mon Mar 15 16:20:42 EST 2021'. The user then runs 'date +%A %B %D', which outputs 'Monday March 03/15/21'. Finally, the user runs 'exit', and the script ends with a timestamp 'Mon 15 Mar 2021 04:21:04 PM EST' and the prompt 'divya@localhost divya\$'.

todos/unix2dos: Converts Linux Files to Windows Format

If you want to share a text file you created on a Linux system with someone on a system running Windows, you need to convert the file before the person on the other system can read it easily. The todos (to DOS; part of the tofrodos package) or unix2dos (UNIX to DOS; part of the unix2dos package) utility converts a Linux text file so it can be read on a Windows system. You can use the fromdos (from DOS; part of the tofrodos package) or dos2unix (DOS to UNIX; part of the dos2unix package) utility to convert Windows files so they can be read on a Linux system.

5.4 Compressing and Archiving Files

Large files use a lot of disk space and take longer than smaller files to transfer from one system to another over a network. If you do not need to look at the contents of a large file often, you may want to save it on a CD, DVD, or another medium and remove it from the hard disk. If you have a continuing need for the file, retrieving a copy from another medium may be inconvenient. To reduce the amount of disk space a file occupies without removing the file, you can compress the file without losing any of the information it holds. Similarly, a single archive of several files packed into a larger file is easier to manipulate, upload, download, and email than multiple files. You may download compressed, archived files from the Internet.

bzip2: Compresses a File

The bzip2 utility compresses a file by analyzing it and recoding it more efficiently. The new version of the file looks completely different. In fact, because the new file contains many nonprinting characters, you cannot view it directly. The -v (verbose) option causes bzip2 to report how much it was able to reduce the size of the file.

```

divya@localhost:~
File Edit View Terminal Go Help
[divya@localhost divya]$ ls -l
total 36
-rw----- 1 divya divya 111 Mar 15 12:26 Count
-rw----- 1 divya divya 111 Mar 15 12:28 Count1
-rw----- 1 divya divya 57 Mar 15 12:38 Days of the week
-rw----- 1 divya divya 130 Mar 15 12:31 LPU
-rw-rw-r-- 1 divya divya 12 Mar 15 16:15 MyFile
-rw-rw-r-- 1 divya divya 74 Mar 15 16:16 MyFile1
-rw----- 1 divya divya 68 Mar 15 12:36 qwerty
-rw----- 1 divya divya 41 Mar 15 12:39 Save
-rw-rw-r-- 1 divya divya 491 Mar 15 16:21 typescript
[divya@localhost divya]$ bzip2 -v typescript
  typescript: 1.860:1, 4.301 bits/byte, 46.23% saved, 491 in, 264 out.
[divya@localhost divya]$ ls -l
total 36
-rw----- 1 divya divya 111 Mar 15 12:26 Count
-rw----- 1 divya divya 111 Mar 15 12:28 Count1
-rw----- 1 divya divya 57 Mar 15 12:38 Days of the week
-rw----- 1 divya divya 130 Mar 15 12:31 LPU
-rw-rw-r-- 1 divya divya 12 Mar 15 16:15 MyFile
-rw-rw-r-- 1 divya divya 74 Mar 15 16:16 MyFile1
-rw----- 1 divya divya 68 Mar 15 12:36 qwerty
-rw----- 1 divya divya 41 Mar 15 12:39 Save
-rw-rw-r-- 1 divya divya 264 Mar 15 16:21 typescript.bz2
[divya@localhost divya]$

```

The bzip2 utility also renamed the file, appending .bz2 to its name. This naming convention reminds you that the file is compressed; you would not want to display or print it, for example, without first decompressing it.

bunzip2 and bzcata: Decompress a File

The bzcata utility displays a file that has been compressed with bzip2. The equivalent of cat for .bz2 files, bzcata decompresses the compressed data and displays the decompressed data. Like cat, bzcata does not change the source file.

```

divya@localhost:~
File Edit View Terminal Go Help
-rw-rw-r-- 1 divya divya 491 Mar 15 16:21 typescript
[divya@localhost divya]$ bzip2 -v typescript
  typescript: 1.860:1, 4.301 bits/byte, 46.23% saved, 491 in, 264 out.
[divya@localhost divya]$ ls -l
total 36
-rw----- 1 divya divya 111 Mar 15 12:26 Count
-rw----- 1 divya divya 111 Mar 15 12:28 Count1
-rw----- 1 divya divya 57 Mar 15 12:38 Days of the week
-rw----- 1 divya divya 130 Mar 15 12:31 LPU
-rw-rw-r-- 1 divya divya 12 Mar 15 16:15 MyFile
-rw-rw-r-- 1 divya divya 74 Mar 15 16:16 MyFile1
-rw----- 1 divya divya 68 Mar 15 12:36 qwerty
-rw----- 1 divya divya 41 Mar 15 12:39 Save
-rw-rw-r-- 1 divya divya 264 Mar 15 16:21 typescript.bz2
[divya@localhost divya]$ bzcata typescript.bz2
Script started on Mon 15 Mar 2021 04:20:15 PM EST
[divya@localhost divya]$ echo The Linux and Shell Scripting
The Linux and Shell Scripting
[divya@localhost divya]$ date
Mon Mar 15 16:20:42 EST 2021
[divya@localhost divya]$ date +"%A %B %D"
Monday March 03/15/21
[divya@localhost divya]$ exit

Script done on Mon 15 Mar 2021 04:21:04 PM EST
[divya@localhost divya]$

```

You can use the bunzip2 utility to restore a file that has been compressed with bzip2.

```
divya@localhost:~
File Edit View Terminal Go Help
[divya@localhost divya]$ ls -l
total 36
-rw-rw-r-- 1 divya divya 111 Mar 15 12:26 Count
-rw-rw-r-- 1 divya divya 111 Mar 15 12:28 Count1
-rw-rw-r-- 1 divya divya 57 Mar 15 12:38 Days of the week
-rw-rw-r-- 1 divya divya 130 Mar 15 12:31 LPU
-rw-rw-r-- 1 divya divya 12 Mar 15 16:15 MyFile
-rw-rw-r-- 1 divya divya 74 Mar 15 16:16 MyFile1
-rw-rw-r-- 1 divya divya 68 Mar 15 12:36 qwerty
-rw-rw-r-- 1 divya divya 41 Mar 15 12:39 Save
-rw-rw-r-- 1 divya divya 264 Mar 15 16:21 typescript.bz2
[divya@localhost divya]$ bunzip2 typescript.bz2
[divya@localhost divya]$ ls -l
total 36
-rw-rw-r-- 1 divya divya 111 Mar 15 12:26 Count
-rw-rw-r-- 1 divya divya 111 Mar 15 12:28 Count1
-rw-rw-r-- 1 divya divya 57 Mar 15 12:38 Days of the week
-rw-rw-r-- 1 divya divya 130 Mar 15 12:31 LPU
-rw-rw-r-- 1 divya divya 12 Mar 15 16:15 MyFile
-rw-rw-r-- 1 divya divya 74 Mar 15 16:16 MyFile1
-rw-rw-r-- 1 divya divya 68 Mar 15 12:36 qwerty
-rw-rw-r-- 1 divya divya 41 Mar 15 12:39 Save
-rw-rw-r-- 1 divya divya 491 Mar 15 16:21 typescript
[divya@localhost divya]$
```

gzip: Compresses a File

The gzip (GNU zip) utility is older and less efficient than bzip2. Its flags and operation are very similar to those of bzip2. A file compressed by gzip is marked by a .gz filename extension. Linux stores manual pages in gzip format to save disk space; likewise, files you download from the Internet are frequently in gzip format. The gunzip utility to restore a file that has been compressed with gzip. zcat decompresses the compressed data and displays the decompressed data.

```
divya@localhost:~
File Edit View Terminal Go Help
[divya@localhost divya]$ ls -l
total 36
-rw-rw-r-- 1 divya divya 111 Mar 15 12:26 Count
-rw-rw-r-- 1 divya divya 111 Mar 15 12:28 Count1
-rw-rw-r-- 1 divya divya 57 Mar 15 12:38 Days of the week
-rw-rw-r-- 1 divya divya 130 Mar 15 12:31 LPU
-rw-rw-r-- 1 divya divya 12 Mar 15 16:15 MyFile
-rw-rw-r-- 1 divya divya 74 Mar 15 16:16 MyFile1
-rw-rw-r-- 1 divya divya 68 Mar 15 12:36 qwerty
-rw-rw-r-- 1 divya divya 41 Mar 15 12:39 Save
-rw-rw-r-- 1 divya divya 491 Mar 15 16:21 typescript
[divya@localhost divya]$ gzip -v typescript
typescript: 57.6% -- replaced with typescript.gz
[divya@localhost divya]$ zcat typescript.gz
Script started on Mon 15 Mar 2021 04:20:15 PM EST
[divya@localhost divya]$ echo The Linux and Shell Scripting
The Linux and Shell Scripting
[divya@localhost divya]$ date
Mon Mar 15 16:20:42 EST 2021
[divya@localhost divya]$ date +"%A %B %D"
Monday March 03/15/21
[divya@localhost divya]$ exit

Script done on Mon 15 Mar 2021 04:21:04 PM EST
[divya@localhost divya]$
```

Linux and Shell Scripting

```

divya@localhost:~
File Edit View Terminal Go Help
[divya@localhost divya]$ gzip -v typescript
typescript: 57.6% -- replaced with typescript.gz
[divya@localhost divya]$ zcat typescript.gz
Script started on Mon 15 Mar 2021 04:20:15 PM EST
[divya@localhost divya]$ echo The Linux and Shell Scripting
The Linux and Shell Scripting
[divya@localhost divya]$ date
Mon Mar 15 16:20:42 EST 2021
[divya@localhost divya]$ date +"%A %B %D"
Monday March 03/15/21
[divya@localhost divya]$ exit

Script done on Mon 15 Mar 2021 04:21:04 PM EST
[divya@localhost divya]$ gunzip typescript.gz
[divya@localhost divya]$ ls -l
total 36
-rw----- 1 divya divya 111 Mar 15 12:26 Count
-rw----- 1 divya divya 111 Mar 15 12:28 Count1
-rw----- 1 divya divya 57 Mar 15 12:38 Days of the week
-rw----- 1 divya divya 130 Mar 15 12:31 LPU
-rw-rw-r-- 1 divya divya 12 Mar 15 16:15 MyFile
-rw-rw-r-- 1 divya divya 74 Mar 15 16:16 MyFile1
-rw----- 1 divya divya 68 Mar 15 12:36 qwerty
-rw----- 1 divya divya 41 Mar 15 12:39 Save
-rw-rw-r-- 1 divya divya 491 Mar 15 16:21 typescript

```

Do not confuse gzip and gunzip with the zip and unzip utilities. These last two are used to pack and unpack zip archives containing several files compressed into a single file that has been imported from or is being exported to a system running Windows.

tar: Packs and Unpacks Archives

The tar utility performs many functions. Its name is short for tape archive, as its original function was to create and read archive and backup tapes. Today it is used to create a single file (called a tar file, archive, or tarball) from multiple files or directory hierarchies and to extract files from a tar file.

```

divya@localhost:~
File Edit View Terminal Go Help
[divya@localhost divya]$ ls -l
total 36
-rw----- 1 divya divya 111 Mar 15 12:26 Count
-rw----- 1 divya divya 111 Mar 15 12:28 Count1
-rw----- 1 divya divya 57 Mar 15 12:38 Days of the week
-rw----- 1 divya divya 130 Mar 15 12:31 LPU
-rw-rw-r-- 1 divya divya 12 Mar 15 16:15 MyFile
-rw-rw-r-- 1 divya divya 74 Mar 15 16:16 MyFile1
-rw----- 1 divya divya 68 Mar 15 12:36 qwerty
-rw----- 1 divya divya 41 Mar 15 12:39 Save
-rw-rw-r-- 1 divya divya 491 Mar 15 16:21 typescript
[divya@localhost divya]$ tar -cvf all.tar typescript LPU Count
typescript
LPU
Count
[divya@localhost divya]$ ls -l all.tar
-rw-rw-r-- 1 divya divya 10240 Mar 15 18:02 all.tar
[divya@localhost divya]$ tar -tvf all.tar
-rw-rw-r-- divya/divya 491 2021-03-15 16:21:04 typescript
-rw----- divya/divya 130 2021-03-15 12:31:39 LPU
-rw----- divya/divya 111 2021-03-15 12:26:21 Count
[divya@localhost divya]$

```

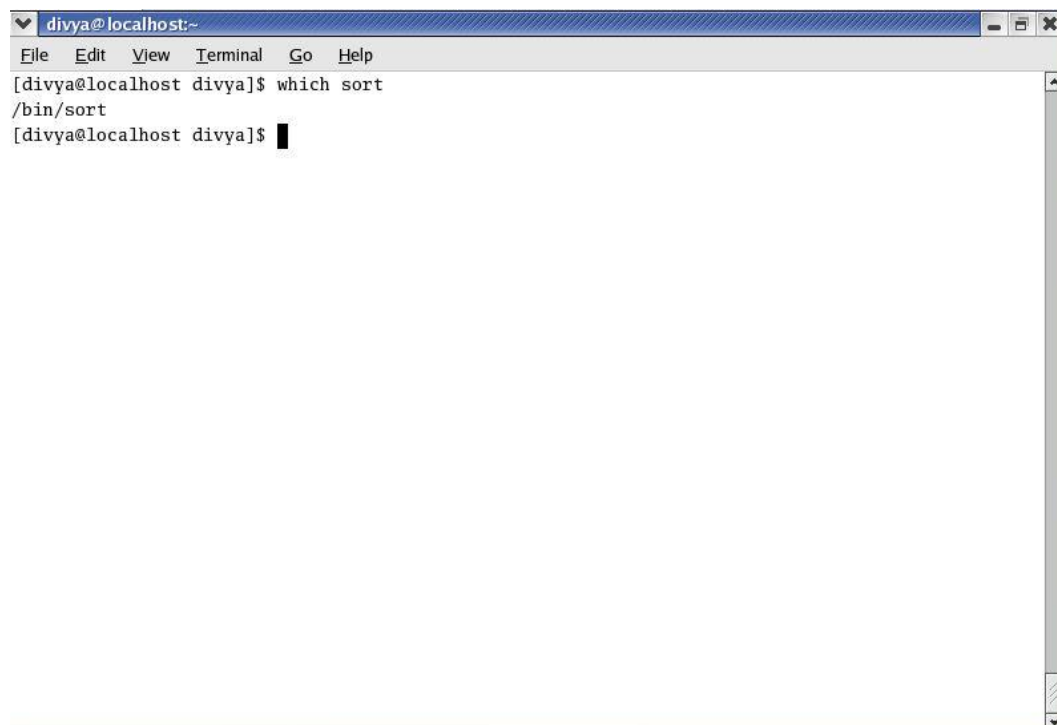

Tar uses the `-c` (create), `-v` (verbose), and `-f` (write to or read from a file) options to create an archive named `all.tar` from these files. Each line of output displays the name of the file tar is appending to the archive it is creating. The tar utility adds overhead when it creates an archive. The final command uses the `-t` option to display a table of contents for the archive. You can use `bzip2`, `compress`, or `gzip` to compress tar files, making them easier to store and handle. Many files you download from the Internet will already be in one of these formats. Files that have been processed by tar and compressed by `bzip2` frequently have a filename extension of `.tar.bz2` or `.tbz`. Those processed by tar and `gzip` have an extension of `.tar.gz`, `.tgz`, or `.gz`, whereas files processed by tar and `compress` use `.tar.Z` as the extension.

5.5 Locating Commands

The `where` and `locate` utilities can help you find a command whose name you have forgotten or whose location you do not know. When multiple copies of a utility or program are present, `which` tells you which copy you will run. The `slocate` utility searches for files on the local system.

Which and where is: Locate a Utility

When you give Linux a command, the shell searches a list of directories for a program with that name and runs the first one it finds. This list of directories is called a search path. If you do not change the search path, the shell searches only a standard set of directories and then stops searching. However, other directories on the system may also contain useful utilities. The `which` utility locates utilities by displaying the full pathname of the file for the utility. The local system may include several utilities that have the same name. When you type the name of a utility, the shell searches for the utility in your search path and runs the first one it finds. You can find out which copy of the utility the shell will run by using `which`.

A screenshot of a terminal window titled 'divya@localhost:~'. The window has a menu bar with 'File', 'Edit', 'View', 'Terminal', 'Go', and 'Help'. The terminal shows the command '[divya@localhost divya]\$ which sort' and its output '/bin/sort'. The prompt '[divya@localhost divya]\$' is followed by a cursor. The terminal window has standard window controls (minimize, maximize, close) in the top right corner and a scrollbar on the right side.

The `where` utility searches for files related to a utility by looking in standard locations instead of using your search path.

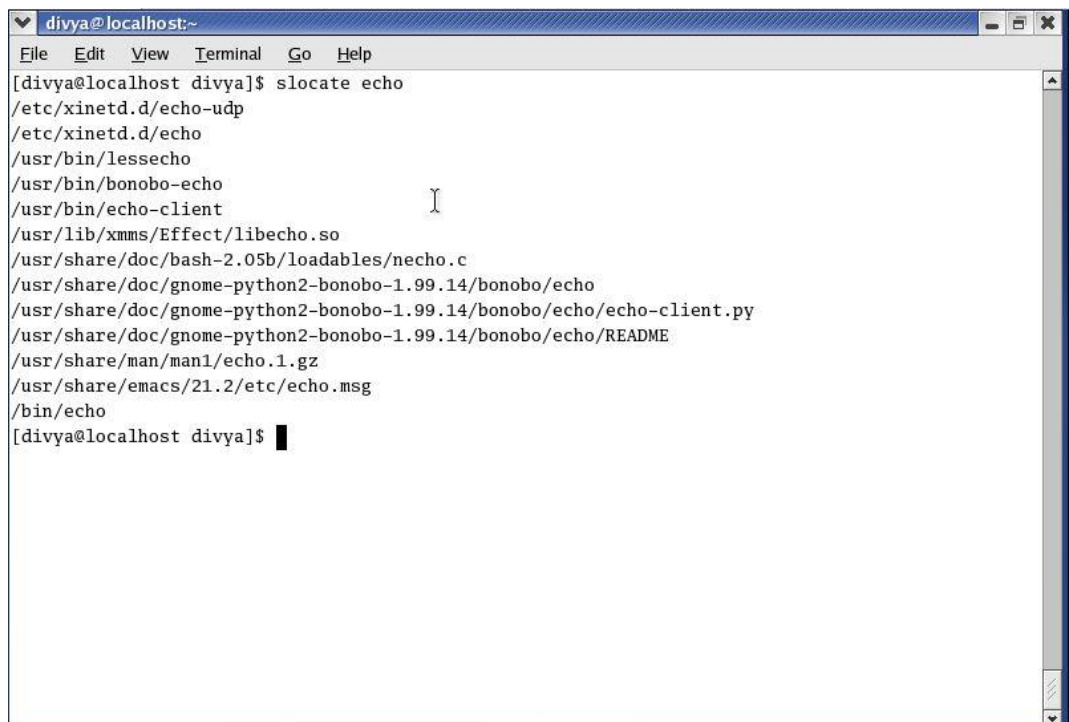


```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ whereis sort  
sort: /bin/sort /usr/share/man/man1/sort.1.gz /usr/share/man/man3/sort.3pm.gz  
[divya@localhost divya]$
```

Where is finds three references to sort: the sort utility file, a sort header file, and the sort man page.

slocate/locate: Searches for a File

The slocate (secure locate) or locate utility searches for files on the local system.



```
divya@localhost:~  
File Edit View Terminal Go Help  
[divya@localhost divya]$ slocate echo  
/etc/xinetd.d/echo-udp  
/etc/xinetd.d/echo  
/usr/bin/lessecho  
/usr/bin/bonobo-echo  
/usr/bin/echo-client  
/usr/lib/xmms/Effect/libecho.so  
/usr/share/doc/bash-2.05b/loadables/necho.c  
/usr/share/doc/gnome-python2-bonobo-1.99.14/bonobo/echo  
/usr/share/doc/gnome-python2-bonobo-1.99.14/bonobo/echo/echo-client.py  
/usr/share/doc/gnome-python2-bonobo-1.99.14/bonobo/echo/README  
/usr/share/man/man1/echo.1.gz  
/usr/share/emacs/21.2/etc/echo.msg  
/bin/echo  
[divya@localhost divya]$
```

Summary:

- When you log in the system, you work in the home directory.

- When you want to view a file that is longer than one screen, you can use either the less utility or the more utility.
- If the destination-file exists before you give a cp command, cp overwrites it. The cp -i (interactive) option prompts you before it overwrites a file.
- If a file contains a list of names and has two successive entries for the same person, uniq skips the duplicate line.
- The diff utility with the -u (unified output format) option first displays two lines indicating which of the files you are comparing will be denoted by a plus sign (+) and which by a minus sign (-).
- You can use the file utility to learn about the contents of a file without having to open and examine the file yourself.
- By default script captures the session in a file named typescript. To specify a different filename, follow the script command with a SPACE and the filename
- The todos (to DOS; part of the tofromdos package) or unix2dos (UNIX to DOS; part of the unix2dos package) utility converts a Linux text file so it can be read on a Windows system.
- Linux stores manual pages in gzip format to save disk space; likewise, files you download from the Internet are frequently in gzip format.
- The gunzip utility to restore a file that has been compressed with gzip.
- The whereis and slocate utilities can help you find a command whose name you have forgotten or whose location you do not know.

Keywords

- **Directory:** A directory is a resource that can hold files. On other operating systems, like Windows, a directory is referred to as a folder.
- **ls:** The ls utility lists the names of files which are available.
- **cat:** The cat utility displays the contents of a text file.
- **rm:** The rm (remove) utility deletes a file.
- **cp::** The cp (copy) utility makes a copy of a file. This utility can copy any file, including text and executable program (binary) files.
- **mv:** The mv (move) utility can rename a file without making a copy of it.
- **grep:** The grep utility searches through one or more files to see whether any contain a specified string of characters.
- **sort:** The sort utility displays the contents of a file in order by lines; it does not change the original file.
- **diff:** The diff (difference) utility compares two files and displays a list of the differences between them.
- **uniq:** The uniq (unique) utility displays a file, skipping adjacent duplicate lines, but does not change the original file.
- **Pipe:** A pipe (written as a vertical bar [|] on the command line and appearing as a solid or broken vertical line on a keyboard) provides the simplest form of this kind of communication.
- **echo:** The echo utility copies the characters you type on the command line after echo to the screen.
- **date:** The date utility displays the current date and time.

- **script:** The script utility records all or part of a login session, including your input and the system's responses.
- **bzip2:** The bzip2 utility compresses a file by analyzing it and recoding it more efficiently.
- **bzcat:** The bzcat utility displays a file that has been compressed with bzip2.
- **bunzip2:** You can use the bunzip2 utility to restore a file that has been compressed with bzip2.
- **whereis:** The whereis utility searches for files related to a utility by looking in standard locations instead of using your search path.
- **slocate:** The slocate (secure locate) or locate utility searches for files on the local system.

Self Assessment

1. Which of these utilities is used to convert a Linux text file so that it can be read on a Windows system?
 - A. todos
 - B. dos2unix
 - C. echo
 - D. script
2. Which of these utilities records a shell session?
 - A. echo
 - B. date
 - C. script
 - D. cat
3. Which of these is not used in Linux System?
 - A. zip
 - B. bzip2
 - C. bunzip2
 - D. zcat
4. Which of these utilities is used to compress a file?
 - A. bzip2
 - B. bunzip2
 - C. gunzip
 - D. zcat
5. Which of these utilities records a shell session?
 - A. echo
 - B. date
 - C. script
 - D. cat
6. In less/more, which of these keys should be pressed to display the next screen?

- A. SPACE
 - B. ENTER
 - C. CTRL
 - D. ALT
7. Which of these utilities removes duplicate lines from a file?
- A. grep
 - B. uniq
 - C. sort
 - D. None of the above
8. Which option with rm provides the interactive deletion?
- A. -i
 - B. -a
 - C. -b
 - D. -r
9. Which of these utilities are used when you want to view a file that is longer than one page?
- A. more
 - B. less
 - C. Both more and less
 - D. None of the above
10. Which of these utility displays the contents of a text file?
- A. ls
 - B. cat
 - C. rm
 - D. All of the above mentioned
11. Which of these utilities displays the names of files that are available?
- A. ls
 - B. cat
 - C. rm
 - D. All of the above mentioned
12. Which of these utilities compares two files and display the difference between them?
- A. grep
 - B. uniq
 - C. diff
 - D. differ
13. Which symbol is used for representation of a pipe?

- A. #
- B. \$
- C. |
- D. &

14. Which of these represents the basic utilities in Linux?

- A. ls
- B. cat
- C. rm
- D. All of the above mentioned

15. When you log in a Linux system, you work in _____ directory

- A. root
- B. home
- C. var
- D. None of the above

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. A | 2. C | 3. A | 4. A | 5. C |
| 6. A | 7. C | 8. A | 9. C | 10. B |
| 11. A | 12. C | 13. C | 14. D | 15. B |

Review Questions:

1. What is command line utilities? Give some examples and explain the usage of few basic utilities.
2. Which basic utility is used to delete a file? How can we make it interactive?
3. What are pager utilities? How these are useful?
4. Which utilities are used to work with files? Explain any five utilities in detail.
5. Explain the use and syntax of echo, date and script utilities.
6. What is compressing and archiving of files? Explain the utilities used for it.
7. Explain what are locating commands?



Further Readings

Mark G. Sobell, A Practical Guide to Fedora and Red Hat Enterprise Linux, Fifth Edition, Pearson Education



Web Links

<https://www.javatpoint.com/linux-commands>